

AI-Enabled Energy Management for Sustainable Smart Greenhouses: An Integrated Review

Soumya Ranjan Biswal^{1,4}, Tanmoy Roy Choudhury², Senior Member, IEEE, Babita Panda¹, and Subhrajyoti Mishra³

¹School of Electrical Engineering, KIIT DU, Bhubaneswar

²Department of Electrical Engineering, National Institute of Technology, Rourkela

³Institute of Agricultural Sciences, Siksha 'O' Anusandhan Deemed to be University, Odisha

⁴Department of AI&DS, SIT College of Engineering, Kolhapur

Corresponding author: Babita Panda (e-mail: babitapfel@kiit.ac.in).

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ABSTRACT Greenhouse based Controlled Environment Agriculture enables year round crop production. Maintaining a specific microclimate requires a significant amount of energy for controlling the atmosphere under the environment. Rising electricity prices, irregular solar energy generation, and specific microclimate requirements create challenges in operations and the economic viability of the system, thus demanding smart energy optimization. This creates a technical need for advanced Demand Side Management (DSM) techniques capable of aligning load operation with forecasted demand, renewable output, and market signals. Traditional strategies such as fixed rules, time slot scheduling, or heuristic controls offer limited adaptability, lack robustness under uncertainty, and do not generalize across climate zones or crop requirements. Recent developments introduce hybrid AI driven DSM frameworks that combine forecasting, optimization, and autonomous control using approaches such as Deep Learning (DL) based predictors, Model Predictive Control schedulers, and exploration Reinforcement Learning policies to enhance renewable utilisation, reduce cost, and maintain microclimate stability. This review systematically analyzes these methodological shifts, comparing forecasting, scheduling, and control approaches in terms of predictive performance, computational overhead, and deployment feasibility. Recurring challenges including limited transferability across climates, insufficient reporting of techno economic outcomes, and weak validation against biological risk constraints are examined alongside potential solutions such as uncertainty quantification, domain adaptation, and assessment of reliability. The outlines research priorities standardised evaluation directions to support robust, scalable, and commercially relevant DSM adoption in smart greenhouse environments.

INDEX TERMS IoT in agriculture, Demand side management, Load forecasting, Energy optimization, Smart Greenhouse

I. INTRODUCTION

The accelerating trend in population growth, urbanization, and climate uncertainty is placing unprecedented pressure on global food systems. By 2037, the global population is projected to approach 9 billion [1], with a majority residing in urban centers. The Food and Agriculture Organization of the United Nations (FAO) has highlighted that meeting the increased demand requires a substantial boost in food production, all while contending with the environmental impacts of traditional agriculture, such as high greenhouse gas (GHG) emissions, water scarcity, and land degradation [2]. This global challenge has catalyzed a search for

innovative and sustainable methods to produce food locally, efficiently, and with a reduced environmental carbon footprint. Traditional food supply chains are often long, complex, and highly susceptible to disruption. The global system relies heavily on transporting food over greater distances, which results in a significant carbon footprint from emissions generated by shipping and trucking [3]. Additionally, this interconnected network is vulnerable to geopolitical tensions, trade restrictions, and climate related events, which can lead to food import reliance and supply chain fragility.

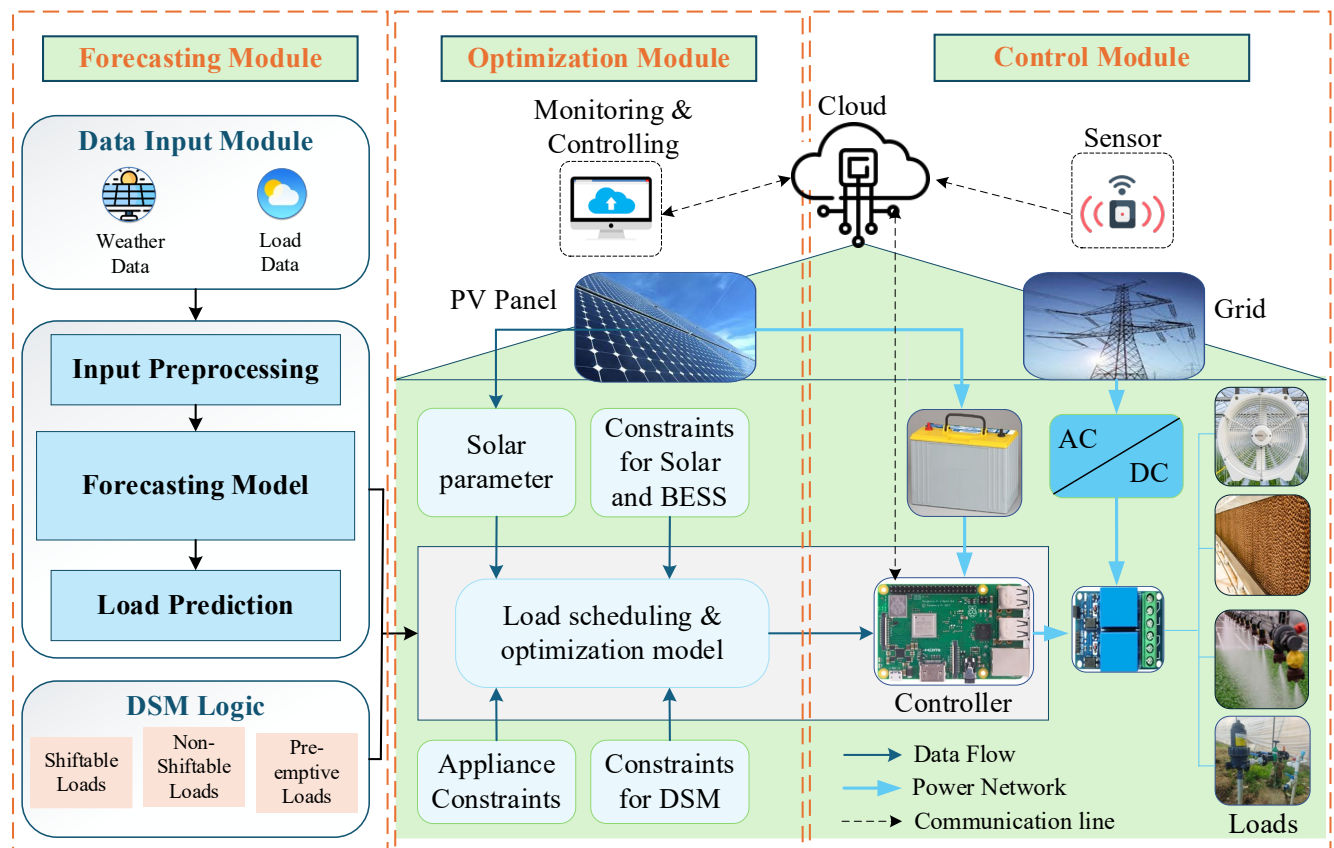


FIGURE 1. Framework of a smart greenhouse energy management

In response to these challenges, Controlled Environment Agriculture (CEA), particularly in the form of greenhouses (GH), has emerged as a promising solution [4]. GH allow for the creation of a favourable microclimate where key environmental factors like temperature, humidity, light, and CO₂ can be precisely controlled. This enables year round production of crops, regardless of climate conditions, and drastically reduces the need for pesticides and water usage compared to open field farming [5], [6]. By offering a way to produce high quality crops in a predictable and consistent manner, CEA represents a crucial step toward localizing food production and minimizing its environmental impact. Fig. 1 illustrates the conceptual framework of a smart greenhouse energy management system. Weather data, including ambient temperature, humidity, wind speed, and solar irradiance, serve as primary exogenous inputs to the forecasting module. Solar irradiance is utilized to predict photovoltaic power generation, whereas other weather variables influence greenhouse thermal and humidity dynamics as well as load demand. Despite their numerous benefits, GH are inherently energy intensive [7]. Maintaining optimal growing conditions, especially in less favourable climates, requires substantial power for heating, cooling, ventilation, and supplemental lighting. This high energy consumption is a major operational bottleneck, driving up costs and contributing to a significant carbon footprint [8], [9]. The challenge is particularly acute during winter and summer

months, when there is an extremely low or high temperature, where the energy required to regulate the internal environment becomes a dominant factor in the overall sustainability and economic viability of the operation.

To mitigate the energy burden, GHs are increasingly integrating renewable energy sources, with solar photovoltaic (PV) systems being a popular choice [10]. Solar PV panels can be installed on rooftops or as dual use canopies, providing a clean source of electricity to power the GH's operations. However, solar energy's intermittency presents a significant challenge, production is dependent on sunlight and doesn't always align with the GH's energy demand [11]. This mismatch between supply and demand necessitates a smarter approach to energy management to fully realize the benefits of renewable integration. This challenge has underscored the need for Demand Side Management (DSM) as a strategic tool for GH energy management. DSM is a set of strategies designed to manage and optimize energy consumption to align with energy availability and cost. In a GH setting, this involves shifting the operation of non critical loads, like irrigation pumps or supplemental lighting, to off peak hours or times of high solar generation [12]. Photovoltaic generation is the most widely adopted renewable source in greenhouse applications due to ease of integration, the proposed framework is not limited to PV alone. Other renewable sources, such as small scale wind turbines, can also be integrated where site conditions permit.

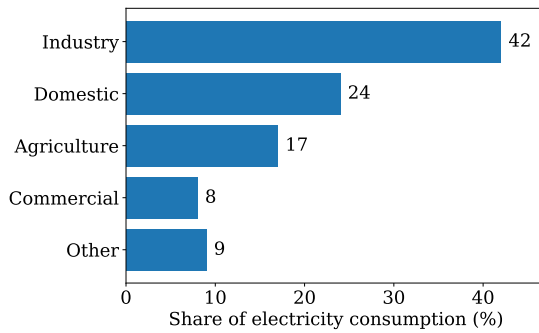


FIGURE 2. Sector wise share of electricity consumption in India

Wind energy can complement solar generation by providing power during low irradiance periods, thereby improving overall renewable availability and system resilience. By integrating DSM, GHs can reduce their electricity bills, increase self consumption of renewable energy, and minimize their reliance on the grid, acting as a crucial bridge between intermittent renewable energy and a constant energy demand [13], [14]. The modern evolution of DSM is being driven by the rise of Artificial Intelligence (AI) and intelligent frameworks [15]. AI based systems use machine learning (ML) to predict energy demand, weather conditions, and electricity prices with greater accuracy [16], [17], [18]. These predictive capabilities, combined with data from Internet of Things (IoT) sensors, enable dynamic and automated control of devices. This allows for sophisticated strategies like predictive control [19], where energy usage is optimized in real time to minimize cost and maximize renewable energy use while maintaining ideal growing conditions [20]. The integration of AI, ML, and IoT thus establishes a new standard for smart, responsive, and efficient GH energy management.

The individual components of GH energy management like forecasting models, load scheduling strategies, control algorithms, and IoT infrastructures have been widely investigated, where as a truly integrated and system level synthesis of AI based DSM remains limited. Existing reviews predominantly examine these elements in isolation, focusing either on hardware and sensing architectures, algorithmic developments, or economic performance, without critically analyzing how their interactions shape overall system effectiveness. This fragmented treatment obscures key cross layer dependencies, the impact of forecasting uncertainty on scheduling feasibility, or the influence of control latency and communication constraints on real time DSM performance. The most DSM research has historically concentrated on residential and industrial energy systems, leaving agricultural applications comparatively underexplored, despite their growing energy intensity. As illustrated in Fig. 2, the agricultural sector alone accounts for nearly 17% of India's total electricity consumption highlighting the strategic importance of greenhouse focused DSM frameworks [21]. Addressing these gaps, this paper advances a pipeline oriented, integrative review of AI based DSM for smart greenhouses, explicitly structured around the

sequential decision layers of the DSM process, forecasting, scheduling and DSM, control, hardware and communication infrastructure, and techno economic evaluation. Rather than treating these components as independent modules, the proposed framework links each method to the operational layer where it acts, allows individual studies to be positioned across multiple dimensions, and elevates techno economic and policy analysis as an explicit part of the DSM pipeline. This pipeline oriented organisation links each method to the decision layer where it operates, allows individual studies to be positioned across multiple dimensions to reveal cross layer gaps, and treats techno economic and policy analysis as an explicit dimension. In this way, the framework extends earlier single axis classifications and supports a more integrated assessment of AI driven DSM in smart sustainable greenhouses.

The main contributions of this review are as follows:

1. A five dimensional, DSM centred classification framework for smart sustainable GH. The framework links forecasting, DSM and load scheduling, control strategies, IoT and communication architectures, and techno economic evaluation as connected stages of one energy management pipeline.
2. A systematic, PRISMA based mapping of studies on AI based DSM in smart GH and CEA. The mapping covers publications between 2015 and 2025 and distinguishes among different system types and DSM objectives.
3. A comparative analysis of modelling and control approaches used in greenhouse DSM, including ML, deep learning (DL), optimization, model predictive control (MPC), and reinforcement learning (RL).
4. A techno economic analysis for AI based DSM in smart GH links technical design choices in DSM, forecasting, and control to economic performance and smart grid participation.

II. METHODOLOGY OF THE REVIEW

The methodology adopted in this review follows a structured approach to ensure the comprehensiveness, and relevance of the collected literature. A multi stage process was implemented, consisting of start with defining the scope and classification dimensions, in 2nd stage performing systematic literature searches across multiple databases, then applying inclusion and exclusion criteria, organizing the results through a PRISMA based flow, and finally identifying research trends. A comprehensive literature search was conducted to capture peer reviewed articles, conference proceedings, and authoritative reports published between January 2015 and March 2025. The selection of this time frame ensures coverage of both foundational works in smart GH energy management and recent advancements in AI driven DSM pipelines. Multiple scholarly databases were consulted to minimize publication bias and ensure coverage across engineering, agricultural, and interdisciplinary domains.

TABLE 1
SUMMARIZES THE TAXONOMY.

Cluster	Representative Keywords	Purpose in Search
Domain Context	Smart Greenhouse, controlled environment agriculture, CEA, protected cultivation	Identify literature specifically on GH or CEA systems
Energy Management	demand side management, DSM, demand response, Energy management, DR, energy optimization, load scheduling	Capture studies focusing on operational strategies for energy use
Forecasting Targets	Load forecasting, energy prediction, solar generation forecasting, renewable generation prediction, weather forecasting, price forecasting	Include works addressing predictive models relevant to DSM pipelines
AI & ML Models	machine learning, ML, deep learning, DL, reinforcement learning, RL, AI, model predictive control, MPC	Retrieve literature applying AI based approaches to GH energy management
IoT & Communication	IoT, Internet of Things, edge computing, cloud computing, digital twin, MQTT, CoAP, ZigBee, LoRa, 5G	Ensure inclusion of studies on hardware, connectivity, and cyber physical integration
Optimization Approaches	Mixed Integer Linear Programming, MILP, Mixed Integer Non Linear Programming, MINLP, metaheuristics, PSO, GA, ACO, multi objective optimization, scheduling algorithms	Target operational optimization frameworks used in DSM
Techno Economic & Policy	Techno economic analysis, cost benefit analysis, ROI, life cycle assessment, LCA, carbon footprint, SDG alignment	Identify studies addressing economic feasibility and environmental impacts
Standards & Interoperability	IEEE 2030.5, OpenADR, AGROVOC, FIWARE, standard protocols	Capture works discussing integration with smart grids and standardized data models

The primary databases included:

- IEEE Xplore: for AI, IoT, control systems, and smart grid integration studies.
- Scopus and Web of Science: for multidisciplinary coverage and citation tracking.
- ScienceDirect and SpringerLink: for energy systems, agricultural engineering, and optimization studies.
- MDPI, Wiley Online Library, and Taylor & Francis: for open access and agricultural sustainability publications.

Google Scholar as a supplementary source for grey literature, theses, and highly cited preprints

A. KEYWORD TAXONOMY AND SEARCH STRINGS

To ensure systematic coverage of the interdisciplinary scope spanning energy forecasting, DSM strategies, control architecture, IoT integration, and techno economic analysis, a structured keyword taxonomy was developed. The taxonomy organized keywords into thematic clusters such as energy forecasting, DSM and optimization, control systems, IoT and sensing, and economic evaluation. This structured approach allowed precise Boolean combinations during database searches, ensuring comprehensive retrieval while avoiding irrelevant results. Table 1 summarizes the keyword clusters and representative search strings used in major databases, which served as the foundation for subsequent literature screening.

B. INCLUSION AND EXCLUSION CRITERIA

A rigorous screening framework was applied to retain only those studies directly contributing to AI driven energy forecasting, DSM, and control strategies in smart sustainable GHs. The framework ensured a balance between technical depth, practical applicability, and interdisciplinary coverage. Table 2 summarizes the inclusion and exclusion parameters,

highlighting domain relevance, methodological requirements, publication scope, and accessibility conditions. These criteria formed the basis for filtering search results and refining the dataset for detailed review.

In the screening process several studies found that combining a greenhouse or controlled environment setting with explicit energy, DSM, or control related content. A greenhouse experiment that reported microclimate and yield data but referred to energy use only in qualitative terms, without modelling electrical loads or quantifying power or cost, were treated as agronomic in focus and therefore not retained. By contrast, greenhouse climate control or MPC studies that provided explicit power or energy indicators were retained and classified primarily under the Control and DSM dimensions. A similar approach was applied to IoT and smart grid literature. IoT based smart farming platforms were only kept when a clearly defined greenhouse or controlled environment module was present and when the platform generated measurable energy, load, control related variables or forecasting of micro environment. Platforms limited to environmental sensing using sensor, decision support, or generic farm management without greenhouse specific energy modelling were excluded as generic smart farming. In smart grid and microgrid DSM studies, attention was given to those cases where the greenhouse load, control strategy, or DSM role was described in some detail, rather than mentioned only as an undifferentiated agricultural load.

C. PRISMA FLOW DIAGRAM FOR STUDY SELECTION

The process of study selection was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines, thereby ensuring methodological transparency, reproducibility, and consistency throughout the review. The PRISMA workflow as shown in Fig. 3 comprised four stages: Identification, Screening, Eligibility, and Inclusion.

TABLE 2
INCLUSION AND EXCLUSION CRITERIA FOR LITERATURE SELECTION

Criterion Type	Description
Inclusion	Domain Relevance - Studies on smart GHs, controlled environment agriculture, protected cultivation systems with explicit energy management, forecasting, load scheduling, and control strategies.
	Optimization Component - Use of AI, ML, DL, optimization algorithms, model predictive control within the DSM pipeline.
	Energy Perspective - Analysis of electrical load profiles, renewable integration, storage management.
	Publication Type - Peer reviewed journal papers, book chapters, authoritative technical reports.
	Publication Date - Published between 2015 and 2025.
	Language - Available in English.
Exclusion	Agronomic Only Focus - Crop physiology, yield, pest management without energy or control aspects.
	Generic Smart Farming - IoT or agriculture applications not specific to GHs.
	Lack of Technical Implementation - Opinion or editorial works without methodology.
	Duplicate Records - Only the most complete peer reviewed version retained.
	Non Accessible Full Text - Unavailable despite open access or institutional search.

- Identification: An initial pool of records was retrieved from the selected databases using the keyword taxonomy defined. The search yielded 1,142 records.
- Screening: After removing duplicates nearly 312 records, the remaining 830 records were screened based on titles and abstracts. At this stage, 458 records were excluded.
- Eligibility: The full texts of the rest 372 records were assessed against the inclusion and exclusion criteria. This step led to the exclusion of 224 records.
- Inclusion: Finally, 148 studies were selected for detailed review, covering all five classification dimensions of the integrated DSM pipeline, i.e. Forecasting, Scheduling or DSM, Control, Hardware, Techno Economics.

The PRISMA framework ensures that the final dataset is comprehensive, high quality, and directly aligned with the objectives of this review.

D. CLASSIFICATION DIMENSIONS

To systematically synthesize the diverse literature, this review adopts a five dimensional classification framework aligned with the sequential stages of an AI driven DSM pipeline for smart GHs as shown in Fig. 4. Each dimension represents a functional stage in the energy management cycle, enabling thematic grouping, structured comparison, and identification of research gaps.

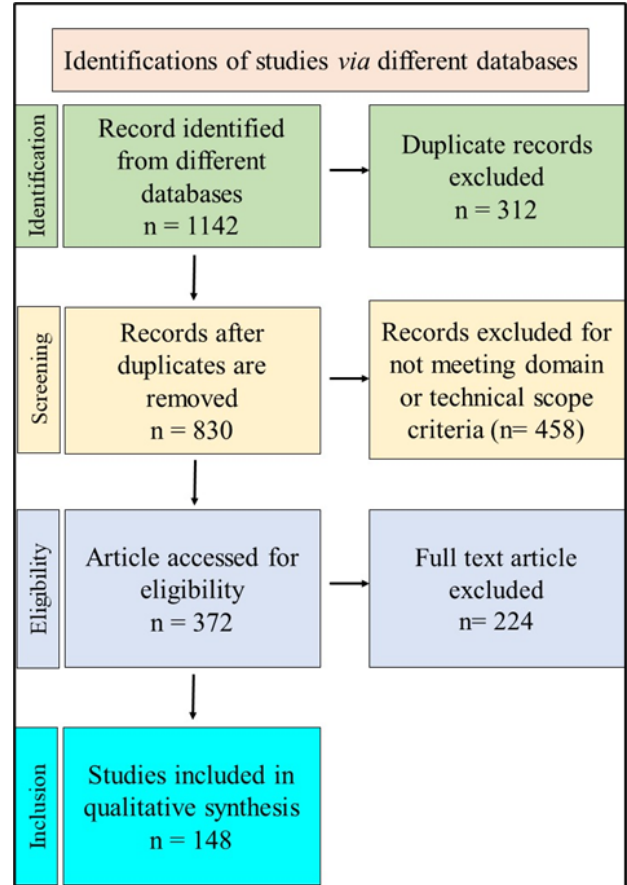


FIGURE 3. PRISMA Flow Diagram

The first dimension, Forecasting, covers the prediction of energy demand, renewable generation, weather variables, and electricity market prices. As the initial stage of the DSM pipeline, it enables proactive control. Studies in this category can be distinguished by their target variables e.g. load, generation, weather, tariff, the type of models employed statistical, ML, DL, or hybrid architectures, the forecasting horizon very short term, short term, or medium term, and the treatment of uncertainty through probabilistic forecasting, confidence intervals, or scenario based methods [22], [23].

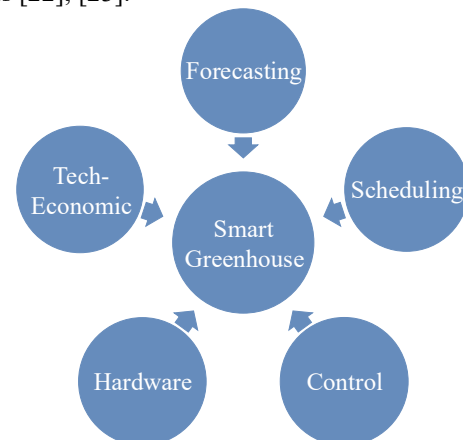


FIGURE 4. Integrative Framework for Smart GH Energy Management

The second dimension, Scheduling and DSM, focuses on optimizing and scheduling GH loads using forecasting outputs under operational and environmental constraints [24], [25], [26]. Approaches range from rule based and priority based methods that offer simplicity and rapid response, to optimization based methods *i.e.*, Mixed Integer Linear Programming (MILP), Mixed Integer Non Linear Programming (MINLP), convex programming, and metaheuristics. MPC enables predictive, constraint sensitive optimization, while RL supports adaptive, self learning scheduling in dynamic conditions [27], [28]. Many studies employ multi objective formulations that balance energy cost reduction, renewable self consumption, and crop comfort [29]. The third dimension, Control, examines the real time execution of DSM schedules and their translation into physical GH actions. Control approaches include conventional methods like ON/OFF, and fuzzy logic, advanced model based control like linear and nonlinear MPC or robust control, and AI driven control derived from RL policies or hybrid AI physics models [30]. Classification often considers the granularity of control and the nature of response. The fourth dimension, Hardware, IoT, and Communication, encompasses the enabling infrastructure for sensing, actuation, and system integration. This includes environmental sensors, smart meters, variable speed drives, IoT gateways and controllers *e.g.*, ESP32, Raspberry Pi, ARM MCUs, and communication protocols such as MQTT, CoAP, ZigBee, LoRa, Wi-Fi, and 5G. Architectural considerations span edge, fog, and cloud computing, often with digital twin (DT) integration for simulation and testing. Cybersecurity measures ensure data integrity and secure control execution [31], [32]. Finally, the fifth dimension, Techno Economic, assesses the viability and sustainability of

AI driven DSM implementations. Economic analysis typically measures energy cost savings, payback period, return on investment (ROI), and benefit cost ratio (BCR) [33]. Environmental impact addresses renewable penetration, and water use efficiency, while LCA examines long term impacts. Socio economic perspectives often emerge from case studies considering policy incentives and adoption barriers [34], [35]. With the five interconnected dimensions, this framework captures both the technological depth and practical relevance concurrently making explicit the dependencies among the different variables. The proposed classification framework is depicted in Fig. 5, providing a structured roadmap for reviewing AI driven DSM approaches in smart sustainable greenhouses. It organizes the literature into three sequential layers forecasting, optimization, and real time control corresponding to how energy decisions are planned, scheduled, and executed. Techniques are further mapped by time horizons, modelling strategies, optimization rules, and control technologies, enabling clear separation between statistical forecasting, mathematical programming, heuristic optimization, and AI based reinforcement learning. This visual layout simplifies comparison across studies and identifies where advanced learning based DSM capabilities are emerging.

III. LOAD PROFILE AND COMPONENTS IN SMART GH

The operation of a smart GH involves diverse electrical loads, each with distinct functions, control characteristics, and flexibility potentials. Classifying these loads is essential for accurate energy modeling, forecasting strategy design, and effective DSM implementation. Table 3 summarizes the major load types based on energy form, primary function, operational characteristics, and flexibility classification.

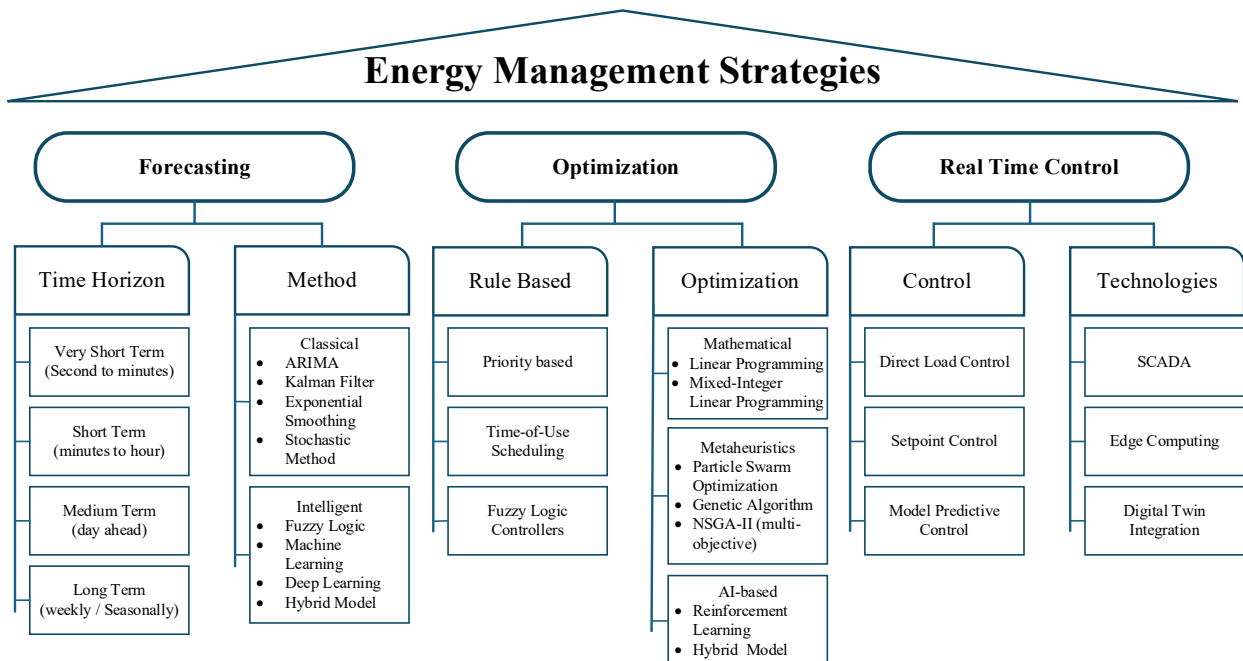


FIGURE 5. Energy Management Strategies in Smart Greenhouses

A. ENERGY CONSUMPTION PATTERNS

The energy profile of a smart GH is dependent on seasonal climate conditions and crop specific microclimate requirements. In winter, heating often represents the largest share of consumption in cold regions, while supplemental lighting becomes dominant in high latitudes during periods of low natural daylight. Summer operation shifts the focus with increased irrigation demand to electrical loads for cooling, ventilation, and fogging, due to enhanced evapotranspiration. Transitional seasons present mixed patterns, with frequent alternation between heating and cooling in the same day, adding complexity to scheduling and control.

Short term weather fluctuations, rapid changes in solar radiation or ambient temperature, further challenge energy management, underscoring the importance of accurate forecasting. Renewable generation interacts strongly with these seasonal patterns. PV systems, the most widely adopted renewable option, align well with summer cooling loads but often mismatch with winter heating requirements, creating seasonal imbalances. Energy storage plays an important role in bridging temporal gaps between supply and demand, battery energy storage systems (BESS) enable shifting of excess PV output to evening or high tariff periods [36]. From a DSM perspective, the interplay between seasonal demand variation, renewable generation profiles, and storage availability defines the scope for cost reduction and carbon mitigation. Aligning flexible loads with renewable peaks can enhance self consumption and reduce grid dependency, but achieving this consistently demands integrated forecasting, scheduling, and control strategies.

The increasing integration of renewable energy sources (RES) into smart GH operations challenges the conventional supply driven grid paradigm, where generation flexibly meets demand [37]. In RES dominated systems, generation is dictated by environmental variability, solar irradiance, wind speed, and temperature necessitating a shift toward demand driven operation. Here, DSM plays a pivotal role, actively reshaping the load curve to enhance system efficiency, reduce peaks, and align consumption with renewable availability [38]. DSM strategies typically include peak clipping, where it limits non essential loads during peak demand, in valley filling, it increases load during low demand periods to stabilize generation and load balance, load shifting relocating operations to off peak times, in conservation overall efficiency improvement, strategic load growth, and flexible load shaping modifying profiles to match RES output.

The economic feasibility of these strategies is often tied to dynamic pricing mechanisms. Time-of-Use (ToU) tariffs that divide the day into on peak and off peak blocks, encouraging load shifting during low tariff periods [39]. Real Time Pricing (RTP) offers finer granularity by updating tariffs in

real time, particularly effective when paired with AI driven predictive scheduling [40]. As shown in Fig. 6 DSM encompasses techniques like peak clipping, valley filling, load shifting, strategic conservation, load growth, and flexible load shaping. Inclining Block Rates (IBR) penalize excessive usage, Day Ahead Pricing (DAP) enables forecast based scheduling, and Critical Peak Pricing (CPP) deters consumption during grid stress events [41], [42].

B. CROP SPECIFIC MICROCLIMATE REQUIREMENTS AND DSM IMPLICATIONS

Greenhouse energy demand and the associated DSM strategies are strongly influenced by crop specific physiological requirements. Different crops exhibit distinct optimal ranges of temperature, relative humidity (RH), CO₂ concentration, and light intensity, which directly determine the operation schedules of heating, cooling, ventilation, lighting, and CO₂ enrichment systems. Consequently, DSM frameworks for smart GH cannot be designed as crop agnostic solutions but must embed crop dependent constraints within forecasting, scheduling, and control layers. For instance, fruiting vegetables as tomatoes and cucumbers typically require higher daytime temperatures and elevated CO₂ levels to maximize photosynthetic activity, resulting in increased heating, ventilation, and CO₂ dosing loads. In contrast, leafy vegetables, lettuce demand lower temperature ranges, moderate humidity, and comparatively lower light intensity, shifting the energy burden toward cooling and ventilation rather than heating or supplemental lighting. These physiological differences translate into distinct load profiles, flexibility margins, and acceptable comfort bands for DSM optimization.

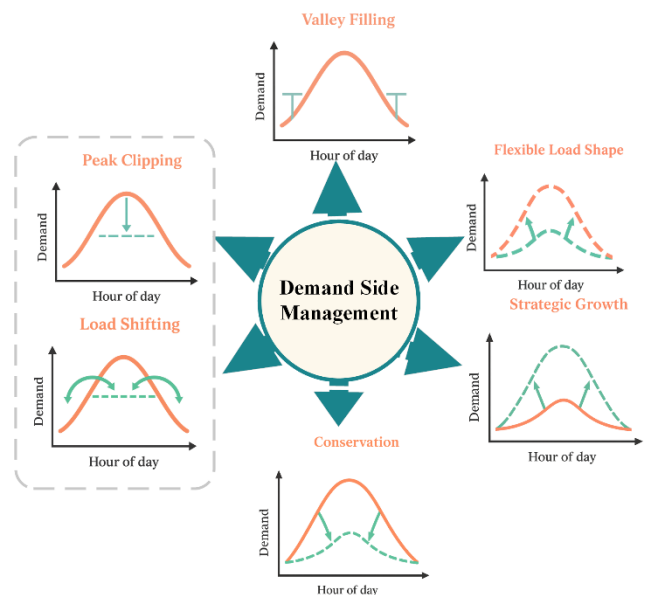


FIGURE 6. Various DSM strategies

TABLE 3
CLASSIFICATION OF ELECTRICAL AND THERMAL LOADS IN SMART GH.

Load Type	Primary Function	Operational Characteristics	Dynamic Response	Flexibility Class
Ventilation Fans	Air exchange and circulation, temperature control during high solar gain	Variable duty cycles; critical in extreme conditions	Fast	Non shiftable / partially shiftable
Foggers & Misters	Humidity increase, evaporative cooling	Low medium power; sensitive to microclimate stability	Medium	Partially shiftable
Irrigation Pumps	Water delivery	Flexible within daily irrigation windows; crop dependent	Slow medium	Fully shiftable
Supplemental Lighting	Photosynthetic light supplementation	High power; shiftable within Daily Light Intensity constraints	Immediate	Partially shiftable
CO ₂ Dosing Systems	CO ₂ enrichment for photosynthesis	Operation linked to lighting schedule	Medium	Partially shiftable
Automation & Control Systems	IoT device operation and actuation control	Low continuous power	Fast	Non shiftable
Cooling Systems	Reduce excess heat in warm seasons	Flexibility depends on thermal inertia and crop tolerance	Medium	Partially shiftable
Dehumidification Units	Prevent fungal diseases	Often coordinated with heating/cooling	Medium	Partially shiftable

In AI driven DSM systems, crop specific requirements are commonly incorporated through constraint boundaries, priority weights, or reward functions. Temperature and humidity setpoints define hard or soft constraints in optimization based DSM and MPC formulations, during which crop sensitivity to microclimate deviations is reflected through penalty terms in multi objective cost functions. In RL based DSM, crop requirements are encoded within the reward structure to penalize actions that violate acceptable comfort zones concurrently encouraging energy efficient operation. This abstraction allows unified DSM architecture to remain crop independent at the framework level while dynamically adapting its operational decisions to crop specific agronomic needs.

Table 4 summarizes representative microclimate requirements for commonly cultivated greenhouse crops, illustrating the variability that DSM strategies must accommodate. The ranges presented are indicative and commonly reported in greenhouse cultivation practice, exact optimal values vary with cultivar, growth stage, and management objectives. The integration of crop sensitive constraints ensures that energy optimization does not compromise crop health, yield, or quality, thereby aligning economic efficiency with agronomic sustainability.

TABLE 4.

TYPICAL MICROCLIMATE REQUIREMENTS OF GREENHOUSE CROPS

Crop	Temperature (°C)	RH (%)	CO ₂ (ppm)	Light Requirement
Tomato	18-26	60-75	800-1200	High
Cucumber	20-28	70-85	800-1000	High
Lettuce	15-22	50-70	400-800	Moderate
Capsicum	18-25	60-70	800-1000	High
Strawberry	16-24	60-80	700-1000	Moderate

C. MATHEMATICAL MODELLING OF GREENHOUSE

Mathematical modelling provides a structured framework to analyze and optimize GH microclimates by capturing heat and mass transfer, radiative exchange, and plant atmosphere

interactions. Coupled with IoT sensing, AI based forecasting, and BESS platforms, these models facilitate predictive control, energy efficiency, and real time decision making [12], [43].

1) TEMPERATURE MODELLING

In smart GH systems, the internal temperature $T_{in}(t)$ plays a pivotal role in determining plant growth and overall productivity. Greenhouse temperature is determined by external weather, structural design, and the operation of cooling and ventilation systems. Developing an accurate model of temperature dynamics is therefore essential for ensuring optimal microclimatic conditions simultaneously reducing energy consumption. The overall thermal behavior of the GH can be expressed as:

$$\frac{dT_{in}(t)}{dt} = Q_{solar}(t) + Q_{vent}(t) - Q_{cooler}(t) - \frac{T_{in}(t) - T_{out}(t)}{R} \quad (1)$$

Where,

$Q_{solar}(t)$ is the heat gain due to solar radiation

$Q_{cooler}(t)$ is the heat removed by the cooling system

$Q_{vent}(t)$ is the heat loss due to ventilation

$T_{out}(t)$ is the internal temperature at time t

R is the thermal resistance between the inside and outside air.

The variation in internal temperature resulting from ventilation or infiltration is primarily governed by the mixing of indoor and outdoor air streams.

The heat exchange due to air exchange is modelled as in equation (2),

$$Q_{vent}(t) = \rho \cdot C_p \cdot V_{air} \cdot \left(\frac{T_{out}(t) - T_{in}(t)}{C} \right) \quad (2)$$

ρ is the air density (kg/m³)

C_p is the specific heat of air (J/kg.°C)

C is the thermal capacity of the GH ($J/^{\circ}C$)

V_{air} is the ventilation airflow rate (m^3/s)

Solar radiation $S(t)$ is a main contributor to the temperature increase in the GH during the daytime. The temperature gain can be define as in equation (22),

$$Q_{solar}(t) = \frac{\alpha \cdot A \cdot S(t)}{C} \quad (3)$$

α is the absorptivity of the GH material

A is the surface area of the GH exposed to solar radiation (m^2)

The operation of cooling system directly affects the internal temperature of the GH. The temperature reduce by the cooler $Q_{cooler}(t)$ is modeled as follows:

$$Q_{cooler}(t) = T_t \cdot \frac{C_{label}(t) \times C_{dur}(t)}{C_{max}(t)} \quad (4)$$

Where, T_t is the max change in temperature due to cooling systems and C_{max} is the maximum operation level and $C_{dur}(t)$ is the interval.

The combined effect of external temperature, solar radiation, and actuator influence on the GHs internal temperature can be described by the following comprehensive model:

$$T_{in}(t) = T_{out}(t) + \frac{1}{C} \left(\alpha \cdot A \cdot S(t) + \rho \cdot C_p \cdot V_{air} \cdot (T_{out}(t) - T_{in}(t)) \right) - T_t \cdot \frac{C_{label}(t) \times C_{dur}(t)}{C_{max}(t)} - \frac{T_{in}(t) - T_{out}(t)}{R} \quad (5)$$

This equation models the internal temperature evolution over time, considering all key factors that influence the GH climate.

2) HUMIDITY MODELLING

In smart GH environments, maintaining appropriate humidity levels is vital for supporting plant physiological processes such as transpiration, photosynthesis, and overall health. Similar to temperature, the internal humidity $H_{in}(t)$ is influenced by external climatic conditions, plant transpiration, and the operation of actuators including fogging systems, humidifiers, and ventilation fans. Precise modelling of humidity dynamics is therefore essential, as it enables the control system to sustain optimal growth conditions concurrently reducing unnecessary energy consumption.

The overall humidity model is represented by a mass balance of water vapor in the air, formulated as:

$$\frac{dH_{in}(t)}{dt} = E_{transp}(t) + E_{vent}(t) + E_{fog}(t) \quad (6)$$

$E_{transp}(t)$ is the rate of water vapor added to air by plant transpiration and soil evaporation, $E_{vent}(t)$ is the rate of water vapor exchange with outside air due to ventilation

(kg/s), $E_{fog}(t)$ is the rate of water vapor added by the fogging system (kg/s)

The present model excludes the effects of transpiration and evaporation on the GH microclimate, as their overall contribution is assumed to be minimal. Similar to thermal dynamics, the internal humidity is affected by air exchange between the GH interior and the external environment through ventilation or infiltration. Accordingly, the rate of water vapor transfer due to this air exchange is expressed as:

$$E_{vent}(t) = \rho_{air} \cdot V_{air} \cdot \left(\frac{H_{out}(t) - H_{in}(t)}{V} \right) \quad (7)$$

ρ_{air} is the air density (kg/m^3)

V_{air} is the ventilation airflow rate (m^3/s)

V is the volume of air in the GH (m^3)

$H_{out}(t)$ is the external humidity

Fogging systems and humidifiers are used to control the internal humidity by adding moisture from the air. The rate of water vapor added by the fogging system is given by:

$$E_{fog}(t) = H_{fog} \cdot \frac{F_{label}(t) \times F_{dur}(t)}{F_{max}(t)} \quad (8)$$

Where, H_{fog} are the max change in humidity due to fogger systems and $F_{max}(t)$ is the maximum operation level and $F_{dur}(t)$ is the interval.

The combined humidity model accounts for the contributions of transpiration, evaporation, fogging, dehumidification, and air exchange. The internal humidity is dynamically adjusted based on these factors, and the overall model can be represented as:

$$H_{in}(t) = H_{out}(t) + \frac{1}{V} \left(\rho_{air} \cdot V_{air} \cdot (H_{out}(t) - H_{in}(t)) \right) + H_{fog} \cdot \frac{F_{label}(t) \times F_{dur}(t)}{F_{max}(t)} \quad (9)$$

This expression models the internal humidity as a function of external conditions, and the operation of the humidity control system. These mathematical formulations provide the foundation for control oriented models, enabling AI driven DSM systems to anticipate seasonal load requirements for heating, ventilation, fogging, and dehumidification. More importantly, the integration of such models with AI and IoT enabled DSM platforms allows smart GH to autonomously schedule loads in accordance with predicted renewable energy availability, forecasted electricity tariffs, and crop specific environmental constraints. This integrated approach not only reduces operating costs but also enhances alignment with a renewable centric grid.

3) LIGHTING LOAD MODELLING

Supplemental lighting is one of the most energy intensive and operationally flexible electrical loads in smart GH, particularly in regions with low natural irradiance or during winter seasons. Artificial lighting systems, typically based on LED or high pressure sodium (HPS) lamps, are used to maintain the required Daily Light Integral (DLI) for optimal crop growth [44], [45]. The electrical power consumption of the lighting system can be modeled as:

$$P_{light}(t) = N_l \cdot P_{rated} \cdot U_l(t) \quad (10)$$

where N_l is the number of lighting fixtures, P_{rated} is the rated power of each fixture, and $U_l(t)$ is a binary or continuous control variable representing the lighting operation level.

The cumulative lighting energy consumption over a scheduling horizon T is given by:

$$E_{light}(t) = \int_0^T P_{light}(t) dt \quad (11)$$

From a plant growth perspective, lighting operation is constrained by the required Daily Light Integral (DLI), which can be expressed as:

$$DLI = \int_{T_{on}}^{T_{off}} PPFD(t) dt \quad (12)$$

Where, $PPFD(t)$ is the photosynthetic photon flux density. In DSM oriented greenhouse operation, artificial lighting is scheduled to supplement natural solar radiation such that the DLI constraint is satisfied and minimizing electricity cost or maximizing renewable energy utilization. Due to its high power rating and temporal flexibility, lighting load is commonly classified as a partially shiftable load and plays a central role in DSM strategies for PV integrated GH.

4) IRRIGATION LOAD MODELLING

Irrigation systems constitute another significant electrical load in smart GH, primarily driven by water pumps, solenoid valves, and fertigation units. Unlike thermal and lighting loads, irrigation demand is event driven and depends on crop type, growth stage, substrate properties, and environmental conditions, like temperature and solar radiation [46], [47]. The electrical power consumption of the irrigation pump can be modeled as:

$$P_{irr}(t) = \frac{\partial_w g Q(t)}{\eta_p} \quad (13)$$

where ρ_w is the water density, g is gravitational acceleration, $Q(t)$ is the volumetric flow rate, H is the total pumping head, and η_p is the pump efficiency.

The total irrigation energy consumption over a period T is given by:

$$E_{irr} = \int_0^T P_{irr}(t) u_{irr}(t) dt \quad (14)$$

where $u_{irr}(t)$ represents the irrigation scheduling signal, typically constrained by crop water requirements and

allowable irrigation windows. In the DSM, irrigation loads are generally classified as fully shiftable within agronomic limits, making them suitable for scheduling during periods of low electricity tariffs or high renewable energy availability. Recent studies demonstrate that coupling irrigation scheduling with energy forecasting can significantly reduce peak demand and operational cost without compromising crop water stress thresholds.

IV. FORECASTING IN SMART GREENHOUSE DSM

Forecasting is the first and most critical stage of AI driven DSM in smart GH, enabling a shift from reactive control to proactive optimization. By accurately predicting short to medium term trends in energy demand, renewable generation, weather conditions, and market tariffs, forecasting models deliver the foundational inputs for load scheduling, shifting, and storage dispatch. Inaccurate predictions risk increasing operational costs, lowering renewable utilization, and compromising crop microclimates. Within an integrated DSM framework, forecasting serves three interlinked roles [48], [49]. Operational planning leverages day ahead forecasts to align shiftable loads with expected renewable peaks or low tariff windows. Real time control benefits from minute to hour horizon forecasts that support rapid adjustments to sudden changes in solar radiation, temperature, or humidity. In grid connected systems, accurate forecasts enable participation in demand response (DR) programs and RTP markets by strategically timing imports and exports. The economic benefits of forecast accuracy are well documented. Case studies show that integrating PV generation forecasts into control reduces operational costs by 5-10% compared to reactive control [50], concurrently multi target models predicting load, PV output, and tariff dynamics achieve up to 12% greater DSM savings in simulation [51].

GH forecasting remains challenging due to strong variable coupling, non linear dynamics, and high sensitivity to climate conditions. Seasonal crop growth changes, stochastic weather, and interdependent thermal electrical loads complicate model generalization. Emerging solutions include hybrid ML models, transfer learning, and probabilistic forecasting. These approaches directly enhance DSM's ability to make cost effective, renewable aligned, and climate resilient decisions. Smart GH DSM, forecasting encompasses several interdependent targets whose accurate prediction directly influences operational efficiency and cost optimization. These targets can be broadly categorized into four domains, energy demand, renewable generation, weather parameters, and tariff or market prices. Energy demand forecasting focuses on predicting short term electrical loads, enabling proactive alignment of consumption with renewable generation availability and prevailing market conditions [52], [53]. Accurate load prediction must incorporate factors with microclimate setpoints, crop growth stages, and operational schedules for lighting, irrigation, and

environmental control systems. Studies have shown that integrating crop specific physiological models within ML frameworks can significantly enhance prediction accuracy compared to purely statistical approaches [54], [55].

Renewable generation forecasting, particularly for on site PV or hybrid renewable systems, is essential for optimizing self consumption and scheduling energy storage[56]. Short term solar output prediction, achieved through DL applied to weather sequences, has been reported to reduce battery cycling frequency and improve renewable energy utilization [57]. Weather parameter forecasting underpins both load and generation predictions by providing localized projections of temperature, humidity, solar radiation, and wind speed. Solar radiation, in particular, serves as a primary driver for PV generation and GH thermal dynamics, simultaneously wind speed influences ventilation and heating requirements. Integration of high resolution numerical weather prediction (NWP) models with AI based post processing has demonstrated improved accuracy in microclimate control strategies for GHs [58]. Tariff and market price forecasting plays a critical role in grid connected scenarios, where DSM algorithms can exploit ToU or RTP variations. Anticipating these trends allows scheduling of flexible loads and battery dispatch during low cost periods or strategic export of excess power during peak price intervals. Recent research has incorporated integrated DL frameworks that embed price forecasts directly into optimal load scheduling policies [59]. Recent advances indicate that multi target forecasting where load, generation, and price signals are predicted jointly offers superior performance over independent models by capturing cross dependencies and shared influencing factors. Nevertheless, the realization of such systems remains challenging due to the need for heterogeneous data integration, robust feature engineering, and the incorporation of uncertainty quantification. Addressing these challenges is essential for the deployment of resilient, region adaptive DSM strategies in smart GHs, as explored in subsequent sections of this review.

A. MACHINE LEARNING MODELS

ML techniques have emerged as powerful tools for GH energy forecasting due to their capacity to model non linear, multivariate relationships between environmental parameters, operational settings, energy consumption or generation profiles. Compared to traditional statistical models, ML approaches can accommodate high dimensional datasets, adapt to diverse operating conditions, and integrate heterogeneous data sources such as numerical weather predictions, in situ sensor measurements, and historical operational records.

Support Vector Machines (SVMs) represent one of the earliest ML models applied to GH load prediction, offering strong generalization capabilities in small to medium sized datasets. Kernel based SVMs, in particular, have demonstrated higher accuracy than linear regression methods

when forecasting short term heating loads under variable weather conditions [60], [61]. However, their scalability is limited in very large datasets or in highly dynamic environment unless supported by feature selection or dimensionality reduction techniques. Tree based ensemble methods like Random Forest (RF) and Gradient Boosting Machines (GBMs) are valued for their robustness to overfitting and ability to model complex feature interactions. RF models have been successfully deployed in GH environments for instance, to predict soil temperature and water content using weather informed models [61]. The artificial neural network (ANN) algorithm, although computationally intensive for real time applications, has been integrated into hybrid frameworks for localized microclimate forecasting [62]. Its interpretability and straightforward implementation make it suitable for small scale deployments, though its inability to capture long term temporal dependencies limits its standalone applicability.

Hybrid ML models a highly optimized gradient boosting framework has proven more accurate than conventional ML techniques with combined with SMA and MFO optimization strategies enhance GH internal conditions [63]. Recent advancements have increasingly combined ML models with feature selection and dimensionality reduction methods such as mutual information analysis, recursive feature elimination (RFE), and genetic algorithms to enhance predictive accuracy while reducing computational overhead, particularly for embedded or edge computing environment [64]. The ML based models establish a strong baseline for GH energy forecasting, their static learning nature can restrict adaptability under evolving operational or environmental conditions. This limitation has spurred a growing shift toward DL architectures and hybrid physical data driven models, which are better suited to capture the temporal dependencies and complex non linear dynamics inherent in GH operations.

B. DEEP LEARNING MODELS

DL techniques have emerged as a powerful alternative to conventional ML approaches for GH forecasting, particularly in applications involving high dimensional, sequential, and strongly non linear time series data. Unlike static ML models, DL architectures can autonomously extract hierarchical features from raw inputs and capture long term temporal dependencies, making them highly suitable for predicting energy demand, renewable generation, and microclimate variables under dynamic environmental and operational conditions.

Long Short Term Memory (LSTM) networks are the most extensively applied DL models in this domain, owing to their ability to overcome the vanishing gradient problem and retain relevant historical context over extended time horizons. Empirical studies have reported that LSTM based models outperform Random Forest and XGBoost by 8-15% in mean absolute error (MAE) for multi hour ahead PV

generation and thermal load forecasts [65], [66], [67]. Their capability to model cyclical patterns for instance diurnal lighting schedules and seasonal load variations has proven particularly advantageous for both demand and generation forecasting. Gated Recurrent Units (GRUs) offer a computationally lighter alternative to LSTMs, achieving similar accuracy with requiring fewer parameters[68]. This reduced complexity makes GRU based architectures well suited for edge computing environment, with reported reductions in training time and inference latency of up to 30% compared to LSTMs, without significant accuracy degradation in day ahead energy demand forecasting [69], [70]. Convolutional Neural Networks (CNNs), though primarily associated with spatial feature extraction in image processing, have been adapted for time series forecasting using one dimensional convolutions. This adaptation allows CNNs to capture localized temporal patterns within weather and operational datasets, thereby enhancing short term forecasting accuracy. Building on this, Transformer based architectures originally developed for natural language processing have more recently been adapted for GH forecasting, where their attention mechanisms facilitate the modeling of long range temporal dependencies and complex feature interactions. [71]. By leveraging self attention mechanisms, transformers can focus on relevant temporal segments without reliance on recurrent connections, enabling efficient modeling of long range dependencies. Initial applications in PV generation and load forecasting indicate that transformer based models can surpass LSTM baselines, particularly in handling irregular sampling intervals and incomplete datasets.

C. HYBRID FORECASTING ARCHITECTURES

Hybrid forecasting architectures combine complementary modeling paradigms often in combination with different models to capture the multi modal, interdependent, and non linear dynamics of GH energy systems. The primary motivation is twofold, leverage the strengths of different model types for distinct feature domains, and enhance robustness across diverse operating conditions, seasonal variations, and heterogeneous datasets.

Model combination hybrids, XGBoost-LSTM or GRU-Random Forest, typically allocate specialized roles to each component. Tree based algorithms excel in modeling static or low dimensional features, like equipment specifications, crop type, or seasonal parameters, due to their strong handling of non linear interactions without temporal dependencies. In contrast, recurrent neural networks (RNNs), LSTM and GRU specialize in sequential data modeling, enabling accurate short and medium term forecasts of weather driven load fluctuations. Notably, in TabNet-LSTM pipelines, TabNet performs feature selection and representation learning from high dimensional tabular inputs, subsequently passing compressed and noise filtered features to an LSTM for temporal prediction, thereby improving generalization and

reducing overfitting. RNN-LSTM architectures remain a dominant choice for GH energy prediction, as they can effectively capture both short and long range temporal dependencies in dynamic environments [72], [73]. When combined with recurrent models in hybrid CNN-LSTM architectures, CNN layers perform high level feature extraction, while the LSTM layers model sequential dependencies, resulting in enhanced forecasting performance [74]. ISGA-AMSCNN-DD combinations, which exploit complementary strengths to capture both spatial correlations and long term temporal dependencies [75].

The hybrid and physics informed architecture consistently outperform single model approaches in accuracy and robustness, they also introduce increased computational overhead, integration complexity, and potential latency. These trade offs are particularly relevant in edge deployed GH controllers, where real time feasibility must be balanced against predictive performance. Emerging research in lightweight hybridization and model compression offers promising pathways to reconcile these competing demands.

D. TRANSFER LEARNING

GH energy forecasting models are often trained on location specific datasets, limiting their applicability to other regions with different climates, crop types, or operational practices. Collecting large volumes of high quality training data for each site is costly and time consuming, making transfer learning (TL) and domain adaptation (DA) attractive strategies for regional model generalization.

Transfer learning involves reusing a pre trained model developed for one GH or climatic zone and fine tuning it with a smaller dataset from the target region [76]. This approach reduces the amount of new data needed and accelerates model deployment. Transfer learning has been successfully applied to GH forecasting, where pre trained DL models were adapted to new crops and climatic conditions using limited local data. In particular, transferred BiLSTM models achieved high accuracy, outperforming raw trained models and confirming the potential of transfer learning to enhance microclimate prediction in data scarce GH environments [77]. Domain adaptation techniques go further by adjusting the feature space or learning process to align data distributions between source and target domains. Approaches like adversarial domain adaptation or feature normalization can correct for differences in weather patterns, daylight hours, or operational schedules. These methods are particularly useful in cross continental applications where climatic and economic conditions differ substantially. A parallel control framework was implemented by integrating MPC with a transferable prediction model, where transfer learning was used to fine tune a pre trained climate model with limited local data, enabling effective GH control across different climatic zones. The approach achieved comparable control accuracy to models trained from scratch simultaneously reducing the data requirement by nearly 80%,

demonstrating its efficiency in adapting to new environments [78]. Despite their promise, TL and DA face challenges in GH DSM applications. Model bias can occur if the source and target domains have fundamentally different control strategies or if local events are not represented in the source data. Additionally, adaptation steps must be computationally efficient to suit edge deployed forecasting systems.

E. COMPARATIVE EVALUATION

Comparative studies in smart GH DSM show that forecasting model performance is highly context dependent, varying with climate, crop type, forecast horizon, and data availability. ML methods, XGBoost and Random Forest generally outperform statistical baselines for short term load and generation forecasting, the DL architectures particularly LSTM and CNN-LSTM achieve lower RMSE for multi hour and day ahead predictions. Hybrid models combining tree based feature extraction with sequence models have demonstrated further error reduction, and physics informed hybrids improve robustness under atypical weather conditions. Forecasting models for smart GH DSM must be evaluated beyond accuracy, considering factors like generalization, adaptability, interpretability, computational complexity, temporal alignment with control cycles, and ease of deployment. High accuracy DL models may struggle with interpretability and real time execution, but simpler statistical models offer transparency and speed but lower predictive performance under dynamic conditions.

The radar chart present in Fig. 7 comparing major forecasting architectures across seven evaluation dimensions relevant to smart greenhouse forecasting, like accuracy, generalization, model complexity, interpretability, temporal sequence fitness, data adaptability, and deployment feasibility. The chart synthesizes qualitative and quantitative evidence reported in recent studies and visually exposes performance asymmetries that are difficult to convey through numerical tables alone. The visualization shows that hybrid deep learning boosting models, particularly XGBoost-LSTM, occupy the dominant area in accuracy, temporal fit, and data adaptability, reflecting their ability to capture nonlinear relationships and long term dependencies. This advantage is accompanied by increased architectural depth, hyper parameter tuning effort, and computational cost, positioning these models toward the higher complexity end of the scale. CNN-LSTM and GRU-XGBoost occupy an intermediate zone, offering balanced accuracy and temporal handling with moderate generalization. Traditional approaches such as Linear Regression and Prophet score lower in accuracy and adaptability but retain favourable scores in interpretability and deployment feasibility, a characteristic that supports rapid implementation in resource constrained greenhouse environments. XGBoost performs competitively on accuracy and interpretability but lacks the stronger temporal sequence representation of recurrent models. By mapping models against multiple dimensions simultaneously, it helps identify

scenario specific trade offs for example, prioritizing interpretability and low cost for small scale greenhouses, or selecting hybrid architectures for precision controlled environments where forecasting accuracy directly influences economic return.

This reinforces the need for balanced approaches, combining predictive strength with operational feasibility, supported by standardized benchmarks for fair comparison. Table 5 summarizes representative forecasting performance metrics reported across the reviewed literature. The absolute values, indicative ranges of RMSE, MAE, and MAPE are reported, acknowledging differences in datasets, forecasting horizons, climatic conditions, and greenhouse configurations. This synthesis highlights the consistent trend that DL and hybrid architectures outperform classical and standalone ML models, particularly for multi horizon and multi target forecasting tasks relevant to DSM decision making.

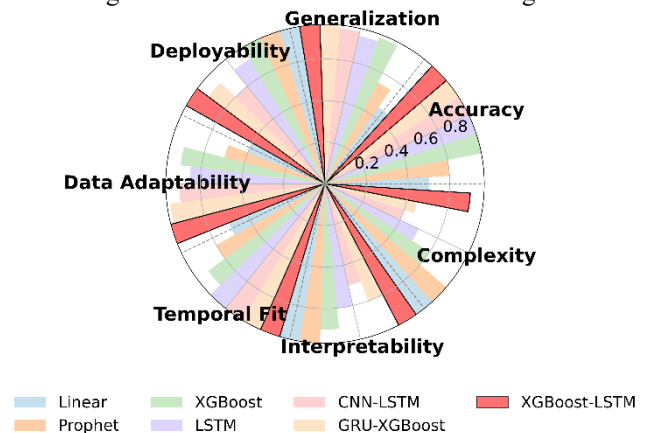


FIGURE 7. Comparing radar chart

V. DEMAND SIDE MANAGEMENT AND LOAD SCHEDULING STRATEGIES

In smart GH operations, DSM serves as the decision making layer that balances economic efficiency with agronomic requirements and renewable energy integration. Unlike conventional building DSM, GH load scheduling must consider the dual objective of cost minimization and crop microclimate preservation, with renewable self consumption emerging as a third priority in hybrid grid PV systems.

Energy Cost Reduction remains the primary driver for DSM adoption, especially under dynamic electricity pricing. Strategies that shift high energy processes like supplemental lighting, irrigation pumping, or CO₂ enrichment towards low tariff or surplus PV generation periods have been shown to reduce operational energy expenditure without compromising production [79].

The economic benefit is amplified in markets with high ToU or RTP volatility, where load flexibility directly translates into avoided peak charges and optimized battery usage. Crop Comfort is equally critical, as deviations from optimal microclimatic setpoints can cause yield loss, delayed growth, or quality degradation. DSM in this context must maintain

TABLE 5.
REPRESENTATIVE FORECASTING PERFORMANCE METRICS REPORTED IN SMART GREENHOUSE DSM LITERATURE

Model Category	Forecast Target	RMSE	MAE	MAPE	
Statistical [80]	Temperature	2.92	2.06	0.11	Suitable for short horizons; limited under nonlinearity
ML (RF, SVR, XGBoost) [81]	Load, PV, Climate	0.52	0.43	1.15	Robust to feature interactions, widely used in GH
DL (LSTM, GRU) [82]	Load, PV, Temp, RH	0.26	0.14	0.004	Captures temporal dependencies effectively
CNN-LSTM / CNN-GRU [83]	Multi target	1.409	1.10	-	Improved short term accuracy via feature extraction
Transformer / Attention Models [84]	Multi horizon	1.350	0.87	0.015	Strong long range dependency modeling
Physics informed / Hybrid AI [85]	Energy	1.61	1.04	-	Improved robustness under unseen conditions

environmental control within narrow thresholds typically $\pm 1-2$ °C for temperature and $\pm 5\%$ RH for humidity and implementing load adjustments. Multi objective formulations are often required, where cost reduction is constrained by environmental comfort penalties to safeguard plant health. Renewable Self Consumption is increasingly prioritized in PV powered GHs, aiming to maximize direct use of on site generation and reduce grid dependence [86]. DSM strategies have demonstrated that aligning GH loads with available solar generation can significantly enhance energy performance. For instance, compared to the fixed load technique, the hybrid energy system (HES) size was reduced by 20.66%, thereby lowering infrastructure requirements. Moreover, by synchronizing lighting, heating, and pumping loads with photovoltaic generation profiles, self consumption ratios increased considerably. This not only decreased reliance on purchased electricity but also mitigated grid export curtailment under restrictive feed in tariffs [87]. The interplay among these objectives introduces trade offs that demand advanced scheduling strategies. Overemphasis on cost reduction, may compromise microclimate stability, at the same time prioritizing comfort without cost consideration can negate DSM's economic value. Optimal DSM design in GHs therefore requires multi objective optimization frameworks capable of dynamically balancing [88], [89].

A. RULE BASED DSM

Early DSM strategies for smart GHs have predominantly relied on deterministic scheduling approaches due to their simplicity, transparency, and low computational requirements. These include rule based, priority based, and time slot based control schemes, each leveraging predefined logic to align load operation with energy availability and cost considerations. In rule based DSM, control actions are triggered when specific environmental or operational

thresholds are reached for example temperature, humidity, or real time energy prices using fixed IF-THEN conditions [90]. It is easy to implement and requiring no training data, where as rule based systems lack adaptability to unforeseen conditions and may lead to suboptimal energy usage during atypical weather patterns.

Priority based DSM ranks load according to their criticality for crop health and production goals. Essential loads e.g. ventilation fans, heating systems receive uninterrupted supply, while non essential or deferrable loads e.g., supplemental lighting, irrigation pumps are curtailed or shifted during peak tariff periods. This approach ensures crop comfort at the same time reducing energy costs, but it relies heavily on accurate load prioritization and may fail to exploit finer cost saving opportunities when priorities are static. Time slot based DSM divides the operational day into predefined intervals, aligning load schedules to exploit low tariff periods. Such methods are effective when paired with TOU pricing schemes and predictable renewable generation patterns, but they may perform poorly under highly volatile RTP or sudden weather fluctuations. Table 6 summarizes the operational characteristics, advantages, and limitations of these three strategies, highlighting their suitability for small to medium scale GHs where computational resources are limited.

B. OPTIMIZATION

Optimization based DSM frameworks aim to determine the most cost effective and energy efficient operating schedule for GH loads whereas satisfying crops requirement and operational constraints. Unlike rule driven approaches, these methods formulate DSM as a mathematical optimization problem, where the objective function typically represents energy cost minimization, renewable self consumption maximization, or a weighted combination of both [91].

TABLE 6
COMPARATIVE SUMMARY OF SIMPLE DSM STRATEGIES

Strategy	Principle	Advantages	Limitations	Study
Rule Based	Fixed control logic from operator rules	Transparent, easy to implement, no training data required	Inflexible, poor performance under rapid changes	Maintaining optimal microclimate for improved plant growth [92]
Priority Based	Assign ranks to loads, allocate power accordingly	Balances cost and crop comfort, suitable for PV battery setups	Requires accurate load classification and thresholds	Priority based DSM with ML predicted sensor data reducing PV storage needs by 63.27% in GH [81]
Time Slot Based	Predefine operation windows based on tariff & forecast	Simple, effective under stable ToU tariffs	Ineffective with RTP or sudden weather changes	ToU based DR in smart grid home achieving 16.5% annual cost savings [93]

Constraints account for environmental setpoints, device specific operational limits, and available renewable generation or storage capacity. MILP is widely used for DSM when both discrete on/off load states and continuous power levels, storage charging rates variables are involved. MILP offers global optimality under linear assumptions and is supported by efficient solvers, making it suitable for day ahead scheduling in GHs with ToU pricing [43].

However, non linearities inherent in complex tariff schemes often necessitate MINLP, which increases computational complexity and may require heuristic or relaxation techniques to find near optimal solutions. Convex optimization methods, where applicable, benefit from strong theoretical guarantees and fast convergence, but their applicability is limited to problems that can be expressed in convex form. When problem complexity or non linear interactions make exact methods intractable, metaheuristic algorithms provide flexible alternatives

PSO has shown strong performance in balancing multiple objectives such as cost, renewable utilization, and peak load reduction, often converging faster than GA for continuous decision variables. GA is effective in combinatorial load

scheduling tasks, especially when constraints involve binary switching and operational sequencing. ACO and DE are less common but have demonstrated competitive results in multi objective DSM scenarios. Hybrid methods, which combine deterministic optimization with metaheuristics, are emerging as a promising approach[94]. MILP can generate a coarse day ahead schedule, which is subsequently fine tuned in real time using PSO or GA to account for updated forecasts of PV output and market prices. Such hybridization offers a balance between computational efficiency and adaptability, making it particularly attractive for semi autonomous GH energy management. Table 7 presents a detailed comparative assessment of various optimization based DSM strategies applied in smart GH environments. The comparison encompasses rule based, mathematical programming, metaheuristic, and hybrid AI optimization approaches, highlighting their underlying algorithms, targeted objectives, key constraints, and operational contexts. This comparative framework serves as a reference point for selecting or developing DSM solutions tailored to specific operational priorities in CEA.

TABLE 7
COMPARATIVE ANALYSIS OF OPTIMIZATION BASED DSM METHODS FOR SMART GREENHOUSES

Method Category	Objective Types	Handling of Uncertainty	Advantages	Limitations	Computational Complexity	Real Time Feasibility	Greenhouse Specific Applications
MINLP [95]	Cost comfort trade off, energy efficiency	Limited direct handling, stochastic variants exist	Models physical processes directly	Computationally expensive, prone to local minima	High (NP-hard)	Low for real time; suited for planning	energy costs, Temperature optimization
Coordinated optimization embedded MPC [96]	Cost, comfort, energy use	Explicit, receding horizon updates	Real time, coordinated control	Needs accurate models	Moderate-High	High	Climate, energy scheduling
Distributionally Robust Optimization [97]	Cost, carbon emission, multi energy scheduling	Explicit, uncertainty sets constructed using Kullback Leibler divergence	Balances robustness and performance	Needs accurate models	High	Low-Moderate	Optimization of energy, water, and heating resources
PSO (Particle Swarm Optimization) [98]	Multi objective	Implicit via randomness	Easy to implement, good for continuous variables	No guarantee of global optimum	Moderate	Good for near real time	photosynthetic rate of crop and energy consumption
GA (Genetic Algorithm) [99]	Multi objective	Implicit parameter variations handled indirectly	Good for nonlinear, combinatorial optimization	Computationally heavy, Convergence speed depend	Moderate-High	Limited for sub minute updates	Temperature and vapor pressure optimization
APSO-GA [100]	Multi objective	Implicit	Improves convergence speed, avoids premature stagnation	Computationally heavy for large problems	Moderate-High	Limited for short term updates	Energy demand, cost
Game-Theoretic Scheduling [101]	Cost sharing, fairness	Implicit	Flexible solver integration	High setup cost	Moderate-High	Moderate	Energy trading in agricultural microgrids
Auction Based Scheduling [102]	Profit maximization, fairness	Implicit	Focuses on upward flexibility	Needs communication	Low-Moderate	High	Bidding for PV energy slots

C. MODEL PREDICTIVE CONTROL (MPC)

MPC has emerged as a prominent technique for DSM in smart GHs due to its ability to optimize control actions over a prediction horizon whereas explicitly considering system constraints. Unlike rule based or purely optimization based strategies, MPC incorporates predictive models of system dynamics where microclimate behavior, load response, and PV generation enabling proactive rather than reactive scheduling.

At each time step, MPC solves an optimization problem using updated forecasts of energy demand, solar generation, weather conditions, and market tariffs, then implements only the first control action before repeating the process in a receding horizon manner. The application of MPC in GH DSM spans a variety of operational domains. Deterministic MPC has been used for short horizon control of lighting and ventilation, aligning operation schedules with forecasted solar generation peaks or low tariff periods.

Stochastic MPC extends this framework by incorporating forecast uncertainty, particularly relevant for renewable rich environment with variable cloud cover or dynamic market prices. Robust MPC further enhances reliability by safeguarding against worst case scenarios, such as sudden weather deterioration, ensuring crop comfort even in extreme events. More recently, learning based MPC approaches integrate online adaptation and RL components to handle non stationary environment, distributed MPC architectures enable scalable multi zone GH control by coordinating local controllers across different subsystems.

To provide a clearer comparison of MPC variants and their applicability in GH DSM, Table 8 summarizes key attributes, benefits, and limitations reported in the literature. MPC provides a flexible and mathematically rigorous framework for GH, DSM, capable of integrating diverse forecasting targets and operational constraints. While deterministic

formulations remain the most computationally tractable, stochastic and robust variants offer improved resilience at the expense of complexity. The ongoing trend toward learning based and distributed MPC architectures aligns with the need for scalable, adaptive, and edge deployable DSM solutions in modern GH systems.

D. REINFORCEMENT LEARNING APPROACHES

RL has gained significant traction in DSM for smart GHs owing to its capacity to learn optimal control policies directly from interactions with the environment, without relying on explicit system models. Unlike MPC, which requires accurate predictive models and repeated optimization at each time step, RL formulates DSM as a sequential decision making problem, where the controller learns to select actions such as load scheduling, storage dispatch, and ventilation control that maximize a long term reward. This reward is typically defined as a weighted combination of energy cost savings, crop comfort preservation, renewable self consumption, and equipment health. Classical RL methods, such as Q learning and SARSA, have been applied to small scale DSM tasks where the state and action spaces are limited, in discrete load switching for irrigation or lighting. Comparative studies in microgrid environments further show that while Q learning and SARSA provide feasible scheduling, deep Q networks (DQN) significantly outperform them achieving up to 12% and 30% higher efficiency, and a 92% improvement over baseline non RL strategies [103]. These results reinforce the need for Deep Reinforcement Learning (DRL) when tackling high dimensional, continuous control problems. DRL extends RL by employing deep neural networks as function approximators, enabling scalable control in complex GH environment.

TABLE 8
COMPARATIVE ANALYSIS OF MPC VARIANTS FOR GREENHOUSE DSM

MPC Variant	Forecast Dependency	Uncertainty Handling	Computational Demand	Typical DSM Applications	Reported Benefits	Limitations
Stochastic MPC [106]	Relies on crop climate model forecasts with parametric uncertainties	Explicitly incorporates probability distributions or scenarios	Moderate	Greenhouse climate control	Provides probabilistic guarantees of constraint satisfaction.	Linearization introduces approximation errors, Performance depends on correct tuning of constraint levels
Data Driven Robust MPC [107]	Relies on weather forecasts, robust with data driven uncertainty	Explicit support vector clustering constructs data driven uncertainty sets	Moderate-High	Greenhouse indoor climate control	14% lower cost, 4% lower cost, Very low constraint violation probability	Requires large historical data sets for uncertainty training
Learning Based MPC [108]	Moderate integrates online learning or RL approximations	Implicit via policy adaptation	Low to Moderate	Embedded DSM for irrigation, ventilation, CO ₂ dosing; adaptive PV integration	Real time feasibility on edge devices; adapts to non stationary conditions	Requires retraining with significant system changes; performance tied to learning quality
Distributed MPC [109]	Depends on subsystem forecast quality	Implicit handled through coordination and scheduling	High	Coordinated energy management in clusters	Improved utilization of RE generation by sharing	Coordination complexity; risk of suboptimal global performance

In DRL based DSM, the agent leverages environmental states comprising real time weather data, PV output, market prices determine control actions that optimize the reward over extended horizons. Beyond generic DRL methods, Ajagekar *et al.* proposed a hybrid framework combining DRL with robust optimization for GH climate control, achieving a 57% reduction in energy consumption and 26.8% improvement in setpoint tracking compared to robust MPC [104]. Importantly, DRL can handle stochastic and non stationary conditions, adapting to changes in weather or crop stages without the need for model recalibration [105].

Recent work has further applied MADRL to networks of renewable integrated GHs, where an actor critic framework with a shared attention mechanism enabled agents to coordinate energy exchanges with the power grid under dynamic pricing and renewable fluctuations. This approach not only maintained indoor climate stability across multiple GHs but also achieved a 28% reduction in net load demand compared with benchmark algorithms [110]. Despite these challenges, some studies have demonstrated the feasibility of integrating RL with IoT technologies in real world GH environments RL based IoT control applied to an industrial hemp production GH successfully maintained target temperatures, reduced energy consumption compared to classical control and minimized the need for continuous human intervention [111]. Safe RL frameworks, which incorporate constraint satisfaction into the learning process, have emerged as a solution to ensure compliance during both training and deployment. In practice, the Soft Action Mask approach has demonstrated this by embedding human defined rules into an RL controller for Venlo GH ventilation and cooling, ensuring safe exploration and optimizing energy use [112].

Advanced coordinated DSM through multi agent and game theoretic frameworks. Evolutionary game theory has been increasingly adopted to model strategic interactions among large populations of energy participants under bounded rationality, offering scalable solutions for decentralized energy trading, demand response, and market based DSM in smart grids [113]. Importantly, recent surveys have highlighted conceptual and mathematical parallels between evolutionary game dynamics and multi agent RL, positioning EGT as a complementary decision making framework for large scale decentralized energy systems. In parallel, multi agent DRL approaches employing centralized training with decentralized execution, shared reward mechanisms, and attention based coordination have demonstrated improved scalability and cooperative behavior in renewable integrated energy systems [114][115]. These developments indicate a clear shift toward coordinated, learning based DSM paradigms capable of handling high dimensional, distributed, and market coupled energy management problems.

RL and its advanced variants offer a model free, adaptive, and potentially more autonomous alternative to optimization based DSM, with the capacity to learn complex control

policies that balance cost efficiency, RE integration, and agronomic constraints. However, their practical adoption will depend on further progress in safe training methodologies, transfer learning techniques for rapid adaptation across facilities, and hardware efficient inference solutions.

VI. CONTROL AND IOT ARCHITECTURES

AI driven energy management in smart sustainable GHs integrates sensing, computation, control, and communication within a unified cyber physical framework. The hardware architecture of the control module is shown in Fig. 8 [81]. The architecture typically follows a layered design that ensures interoperability, modularity, and real time responsiveness across the entire Forecasting, DSM, Control pipeline [116], [117]. At the foundational level, the sensing layer comprises environmental sensors for temperature, humidity, CO₂ concentration, solar radiation, light intensity, and wind parameters, along with smart meters for monitoring electrical loads, renewable generation, and battery state of charge. Continuous acquisition of actuator state variables such as the operating status of ventilation fans, foggers, irrigation valves, shading systems, and lighting units ensures that the control system receives accurate real time feedback. Sensor redundancy, fault detection, and periodic calibration are essential in this layer to maintain reliability under the high humidity and dust prone conditions typical of GH environment [118]. Above this, the network layer facilitates data exchange between field devices, local controllers, and cloud or edge processing units. Communication technologies are selected based on requirements for range, data rate, latency tolerance, and power consumption, with protocols such as MQTT, ZigBee, LoRaWAN, Wi-Fi, and emerging 5G solutions commonly employed. Hybrid networking is often adopted, wherein low power wireless sensor networks communicate with IoT gateways that provide IP based backhaul to higher level computing platforms [119].

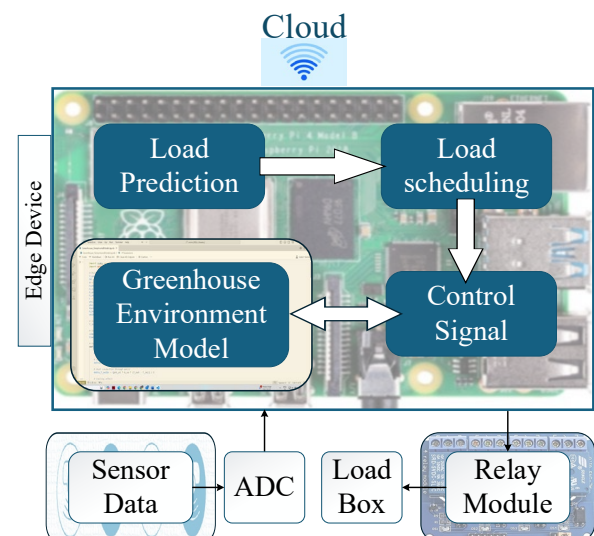


FIGURE 8. Block diagram of the control module.

TABLE 9
COMPARATIVE FEATURES OF COMMUNICATION PROTOCOLS FOR SMART GREENHOUSE DSM

Protocol	Communication Model	Range	Data Rate	Power Consumption	Latency	Suitability in DSM Pipeline
MQTT	Publish-Subscribe	Depends on network (Wi-Fi/Cellular)	High (depends on transport)	Low-Moderate	Low	Real time and cloud integration
CoAP	Request-Response	IP based (LAN/WAN)	Moderate	Low	Low	Lightweight sensor data exchange
ZigBee	Mesh Networking	10-100 m	250 kbps	Very Low	Low	In GH sensor networks
LoRa/LoRaWAN	Star Topology	Up to 15 km	0.3-50 kbps	Very Low	High	Inter GH cluster communication
Wi-Fi	Point to Point / Infrastructure	50-100 m	Up to 600 Mbps	High	Low	High data rate applications
5G	Cellular	>1 km (cell coverage)	Up to 10 Gbps	Moderate	Ultra Low	Time critical DSM and VPP integration

The choice of communication protocol significantly affects performance, reliability, and scalability. Table 9 summarizes the key characteristics of widely adopted protocols, highlighting their operational ranges, power requirements, and integration potential within the DSM pipeline. The processing layer forms the computational core of architecture. Edge computing units execute short term forecasting, DSM algorithms, and local optimization tasks. Cloud platforms support long term analytics, historical data management, and retraining of AI models. In advanced implementations, a BESS of the GH operates in parallel with the physical system, enabling real time simulation of environmental and energy scenarios, testing of control strategies, and safe deployment of algorithm updates. The decision making logic at this layer translates predictive model outputs into control actions that simultaneously optimize energy efficiency, maintain crop comfort, and respect operational constraints. Edge computing platforms in smart GH, DSM architectures enable low latency decision making and reduced reliance on constant cloud connectivity. Device selection depends on computational requirements, power availability, AI processing needs, and integration with IoT communication protocols. Table 10 summarizes key specifications and suitability of hardware platforms for DSM applications. The architectural choice depends on application requirements, cost constraints, and scalability targets, with hybrid models increasingly preferred for robust DSM deployment.

A. DIGITAL TWIN INTEGRATION FOR SIMULATION

A DT is a virtual representation of a physical GH that mirrors its structural layout, environmental conditions, energy flows, and operational dynamics in real time as shown in Fig. 9. In the context of AI driven DSM, DTs provide a safe and cost effective platform for simulation, optimization, and predictive control testing before field deployment [120]. They enable the continuous synchronization of physical sensor data with a virtual model, allowing stakeholders to observe the impact of scheduling, load control, and renewable integration strategies under different climatic and market scenarios [121].

From an architectural perspective, GH DTs integrate IoT enabled sensing networks, edge computing units, and cloud based analytical engines to update the virtual model in near real time. These models can incorporate both energy system components and biological subsystems, enabling a holistic representation of GH performance [122], [123]. Recent work has extended DT architecture by exploring advanced communication systems in GH and underground environments. A novel DT framework employing Visible Light Communication (VLC), alongside WiFi, mobile, and Free Space Optics (FSO) networks, was validated for reliable data exchange. In GH experiments, VLC adjustments improved signal quality by nearly 10 dB over a 4 m distance, demonstrating the potential of DT enabled communication infrastructures for robust data synchronization with control rooms [124]. Several studies have demonstrated DT applications for multi objective optimization in agriculture, where predictive models for weather, market prices, and plant physiology are integrated into the DSM decision making process [125]. The adoption of DTs in smart GHs is still emerging, with key research challenges including data interoperability, model scalability across different GH types, and integration of high fidelity crop models with real time control loops.

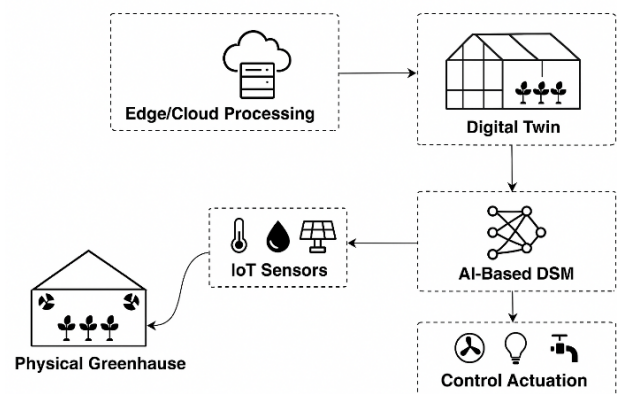


FIGURE 9. Digital twin framework for smart greenhouse.

TABLE 10
COMPARATIVE FEATURES OF HARDWARE PLATFORMS FOR SMART GREENHOUSE DSM

Platform	Processing Capability	AI/ML Support	Power Consumption	Connectivity Options	Typical Cost (USD)	Suitability in DSM Pipeline
ESP32	Dual core 240 MHz MCU	Limited (TinyML)	Very Low (<1 W)	Wi-Fi, Bluetooth	<10	Lightweight sensing & control tasks
Raspberry Pi 4	Quad core 1.5 GHz CPU	Moderate (TensorFlow Lite, OpenCV)	Low-Moderate (3-7 W)	Ethernet, Wi-Fi, Bluetooth	35-75	Forecasting, local optimization
NVIDIA Jetson Nano	Quad core ARM Cortex A57 + 128 core GPU	High (Deep Learning, CUDA)	Moderate-High (5-10 W)	Ethernet, Wi-Fi (via adapter)	99-129	Real time AI inference, vision based control
ARM based MCUs	Varies (e.g., STM32 series)	Limited (TinyML)	Very Low (<1 W)	UART, SPI, I ² C, LoRa, ZigBee	5-20	Ultra low power sensing/control

Nevertheless, DT technology is poised to become a cornerstone of autonomous, self optimising GHs, serving as the experimental backbone for AI driven forecasting, DSM, and adaptive control strategies envisioned in future.

B. PRACTICAL DEPLOYMENT CHALLENGES

The optimization and control algorithms such as DSM, MPC, and RL represent the decision making core of smart greenhouse energy management, their effectiveness is fundamentally constrained by the quality, reliability, and availability of sensor data. Greenhouse environments are inherently harsh for electronic sensing systems due to high humidity, temperature fluctuations, dust, chemical exposure from fertilizers and pesticides, and prolonged operation cycles. These conditions accelerate sensor drift, calibration loss, intermittent failure, and communication dropouts, directly affecting the accuracy of environmental and energy measurements. Sensor drift in temperature, humidity, CO₂, and solar radiation measurements introduces systematic bias that can propagate through forecasting models and subsequently degrade DSM performance [126]. Studies have shown that even small measurement deviations can lead to incorrect load scheduling decisions, resulting in unnecessary energy consumption or violations of crop comfort constraints [127]. Despite this, many AI based DSM studies assume ideal sensor behavior and neglect long term degradation effects, limiting the reproducibility and real world robustness of proposed solutions.

Data availability and labeling present an additional bottleneck for AI driven approaches. High quality, well labeled datasets covering multiple seasons, crop cycles, and operational modes are expensive and time consuming to obtain. Manual labeling of fault conditions, load states, or abnormal microclimate events requires expert intervention, making large scale dataset creation economically challenging [128], [129]. As a result, many studies rely on short term datasets or simulated data, which may not capture rare but critical operating scenarios such as sensor faults, extreme weather events, or equipment failures. The decisive influence of data quality on higher level algorithms is particularly evident in learning based methods. Deep learning and reinforcement learning models are highly sensitive to noise, missing values, and biased samples, often exhibiting

performance degradation or unsafe control actions when deployed under data distribution shifts [130]. In contrast, simpler rule based or physics informed controllers may offer greater robustness under imperfect sensing conditions, highlighting the need for hybrid architectures that explicitly account for data uncertainty.

Emerging solutions to address these challenges include sensor redundancy, online fault detection and diagnosis, adaptive calibration techniques, data driven anomaly detection, and the integration of soft sensors or virtual sensing frameworks [131]. Additionally, transfer learning, self supervised learning, and digital twin assisted data augmentation are gaining attention as means to reduce data dependency by maintaining model generalization. Explicit consideration of sensing reliability and data integrity is therefore essential for transitioning AI based DSM frameworks from experimental validation to scalable, real world greenhouse deployment.

VII. TECHNO ECONOMIC EVALUATION

The techno economic viability of AI driven DSM in smart GHs is commonly assessed using quantitative indicators that capture cost efficiency, financial returns, and operational performance. Energy cost quantifies operational expenditure on electricity over a defined period, incorporating dynamic tariffs such as ToU or RTP. Effective DSM strategies minimize this cost through load shifting, enhanced renewable self consumption, and participation in demand response (DR) programs, in PV integrated GHs, optimisation of time shifting loads has been shown to maximize income and profit while ensuring feasible operation [86]. Multi agent AI controllers for networked GHs under dynamic pricing have reported ~28% net load reductions with maintaining crop comfort, underscoring the cost saving potential of scalable AI DSM [110]. By adopting IoT based optimizations in smart GHs can yield significant economic benefits. An IoT driven optimization scheme for GH climate control reduced energy consumption by 26.56% and cut operating costs by 27.76% compared to a baseline approach [132]. This highlights how automation directly translate into cost savings, thereby improving the economic efficiency of GH operations.

The payback period, expressed over the years, reflects the time required for cumulative energy savings to recover

capital investments in DSM hardware, software, and integration. In typical greenhouse operations without demand side optimization or policy incentives, payback periods for energy related investments often exceed 8-10 years. However, with the implementation of AI driven DSM strategies and the availability of favorable tariff differentials and policy incentives, several studies have reported significantly reduced payback periods ranging from 3 to 7 [86], [110]. ROI, given as a percentage, evaluates the net financial gain relative to initial expenditure, real world IoT based hybrid forecasting and DSM platforms have achieved 17% electricity cost reduction and 7% lower grid dependency, demonstrating practical ROI improvements in operational settings [12]. Similarly, the BCR compares the present value of benefits to costs, and when combined with net present value (NPV) and internal rate of return (IRR) provides a long term profitability profile under variable weather, tariffs, and renewable availability [49], [133]. Recent studies also highlight how tariff design itself critically affects fairness and efficiency. Analysis of tariff structures in a renewable dominated grid context showed that traditional flat rate tariffs can create wealth transfers non owners of distributed renewables pay 22% more than their fair share and time based tariffs achieve both greater economic efficiency and fairness [134]. These findings underscore that tariff reform is not only important for household level equity but also directly impacts GH operators' viability under DSM frameworks.

GH operating costs are dominated by labour and energy, even modest controller improvements yield significant economic benefits. Studies show that labour and energy comprise more than 50% of GH costs, and AI IoT monitoring reduces reliance on costly skilled operators, delivering direct socio economic advantages alongside energy savings [81]. Sensitivity analyses are further applied to capture uncertainties in future energy prices, equipment degradation, and crop yield responses, with some works integrating carbon credit revenues and water savings for full sustainability assessment [49]. Beyond short term cash flow metrics, life cycle cost (LCC) assessments offer a more holistic view of system viability. A recent comparative LCC study of four solar based integrated GH systems including PV, PV/T with nanofluids, desalination, and advanced cooling demonstrated that design innovations can reduce life cycle costs by up to 22.3%. Systems employing spectrum selective nanofluid achieved lower cooling loads and more efficient desalination preheating, yielding leveled costs as low as 0.033 \$/kWh for electricity and 0.015 \$/kWh for cooling [133]. These findings highlight how system level design choices can drastically influence cost effectiveness over the lifetime of GH infrastructure. AI driven DSM also advances environmental sustainability by increasing renewable self consumption and lowering reliance on fossil fuel based grid electricity, thus reducing CO₂ emissions [110]. DSM enabled irrigation and fogging, when combined

with weather or evapotranspiration forecasts, also enhance water use efficiency, improving the resource productivity of smart GHs. Although not a primary performance metric, LCA enriches techno economic evaluations by capturing the environmental footprint of DSM enabled systems from equipment manufacturing to disposal. Findings indicate that the largest environmental gains arise from reduced operational energy use and enhanced renewable integration, which offset embodied impacts of control hardware and storage [34], [49]. Incorporating LCA and LCC into techno economic assessments thus supports policy design and technology selection toward greener and economically viable GH operations.

A. Socio Economic

Socio economic case studies show that the success of AI driven DSM in GHs depends on local tariffs, climate, capital costs, and policies. In Europe, high electricity prices, carbon policies, and demand response markets make DSM with PV and small batteries highly profitable, often achieving payback within 4-6 years when paired with self consumption strategies and storage, especially under dynamic tariffs and carbon pricing regimes [135].

In China, large scale PV GH clusters and strong peak valley tariff differentials make aggregator led load shifting cost effective, with case studies reporting annual returns of 9-20 % and payback periods of 4 to 8 years, alongside social benefits such as job creation and CO₂ reduction [136]. In India, low cost edge devices combined with PV self consumption have emerged as attractive solutions for small and medium farmers, although viability still depends on tariff reforms, reliable metering, and service based O&M models [137]. In North America, demand charges and winter heating loads dominate, with DSM being most effective when integrated with demand response programs via aggregators [138]. Across regions, three common insights emerge. First, supportive policies such as clear rules for exporting surplus solar power and well defined tariff mechanisms are critical for ensuring favorable economic outcomes. Second, service based business models that shift costs from large upfront investments to affordable ongoing payments make adoption easier and less risky for farmers. Third, human in the loop DSM that uses explainable AI and transparent dashboards helps maintain trust, allowing cost savings to be achieved while still protecting crop comfort and yields.

Techno economic assessments demonstrate that the effectiveness of AI based DSM in smart GH is strongly influenced by regional factors such as energy tariffs, policy incentives, climate conditions, and infrastructure maturity. European studies report significant cost reductions driven by high electricity prices and carbon policies, often achieving payback periods below six years. In China, large scale greenhouse deployments benefit from economies of scale and centralized optimization, resulting in improved benefit cost ratios and reduced grid dependency. In African and

other developing regions, techno economic evaluations emphasize affordability, off grid resilience, and socio economic benefits, where even modest energy savings translate into substantial improvements in reliability and food security. These regional differences highlight the broad applicability of DSM principles and underscoring the need for context specific system design. A comparative summary of representative regional techno economic characteristics and reported outcomes is presented in Table 11, highlighting how DSM objectives and economic benefits vary across geographic and socio economic contexts.

TABLE 11.
REGIONAL TECHNO ECONOMIC CHARACTERISTICS OF SMART GREENHOUSE DSM

Region	Energy Context	DSM Objective	Economic Outcome
Europe [139],[140]	High tariffs, carbon pricing	Cost & emission reduction	15-30% energy cost savings; 3-6 yr payback
China [141], [142]	Large scale PV integration	Grid independence	Up to 40% grid import reduction
Africa [143], [144]	Off grid / weak grid	Reliability & affordability	Reduced diesel use; improved ROI
Middle East [144]	Cooling dominated loads	Energy efficiency	Reduced cooling cost; PV viability

B. Environmental and Sustainability

The economic indicators, energy cost savings, payback period, and return on investment remain central to the assessment of DSM strategies in smart GH, environmental and sustainability metrics are increasingly recognized as equally important evaluation dimensions. AI based DSM frameworks influence not only operational expenditure but also long term environmental performance by shaping energy sourcing, load scheduling, and resource utilization patterns.

Life cycle assessment (LCA) has emerged as a key tool for evaluating the environmental footprint of greenhouse energy systems beyond operational phases [145]. Studies integrating AI based scheduling with renewable energy systems report reductions in life cycle greenhouse gas emissions by minimizing reliance on fossil fuel based grid electricity and improving renewable self consumption [146][147]. Although full LCAs are often conducted at the system design stage, DSM decisions significantly affect use phase emissions, which typically dominate the life cycle impact of CEA. CO₂ emission reduction is another critical sustainability indicator directly linked to DSM performance. By shifting energy intensive loads such as supplemental lighting, heating, and irrigation pumping toward periods of high photovoltaic availability or low carbon grid supply, AI driven DSM strategies reduce indirect CO₂ emissions associated with electricity consumption [148]. In grid connected GH, predictive DSM combined with tariff and carbon conscious

scheduling has been shown to lower emission intensity at the same time maintaining crop comfort constraints.

Water savings represent an important but often underemphasized sustainability benefit of intelligent DSM. Energy responsive scheduling of irrigation and fertigation systems, when coordinated with microclimate forecasting, enables precise water delivery aligned with crop demand and evapotranspiration dynamics [149]. Reduced over irrigation not only conserves water but also lowers the energy required for pumping and treatment, reinforcing the water energy nexus in smart greenhouse operation. Beyond individual indicators, sustainability oriented DSM frameworks increasingly adopt multi objective evaluation, where economic cost, energy efficiency, carbon emissions, and water use efficiency are jointly optimized. This integrated perspective aligns AI based DSM with broader sustainability goals and policy objectives, positioning smart GH as active contributors to low carbon, resource efficient agricultural systems rather than isolated energy consumers.

VIII. Standardization and Interoperability

A critical challenge impeding the widespread deployment of integrated AI driven energy management in smart GHs is the lack of standardization and interoperability across IoT hardware and software platforms. Without a robust interoperability framework, valuable data remains trapped in proprietary silos, hindering the development of sophisticated AI models that require a holistic view of both internal microclimate conditions and external grid signals. By adopting open standards, GH operators can integrate devices from multiple manufacturers, enable plug and play deployment, and reduce system integration costs [150]. As the GH sector increasingly embraces DTs, edge AI, and blockchain based energy markets, the role of interoperability standards will be central to achieving reliable, scalable, and secure DSM solutions.

Standardization facilitates consistent data representation, allowing AI algorithms to process, share, and interpret information across platforms without extensive customisation. Different standards play complementary roles some define communication with utilities, others enable grid edge coordination, device level interoperability, or uniform environmental data models. A summary of these standards and their relevance to GH DSM is presented in Table 12. Together, they form the foundation for seamless DSM implementation in smart GHs. By aligning DSM systems with these standards, GH operators can ensure interoperability, regulatory compliance, and scalability. More importantly, real world demonstrations confirm the practical value of these standards in GH DSM. For instance, IEEE 1451 based smart transducer interfaces have been employed to implement a plug and play temperature monitoring and airflow control system in GHs, where distributed sensors automatically integrated into the network enabled real time balancing of internal climate conditions [151].

TABLE 12
SUMMARIES SELECTED STANDARDS WITH RELEVANCE TO AI DRIVEN DSM PIPELINES:

Standard	Domain	Description	Key Relevance to Greenhouse DSM
IEEE 2030.5 (Smart Energy Profile 2.0) [150]	Smart Grid Communication	IP/HTTP protocol for smart energy, covering DR, pricing, and DER coordination	Supports DR program participation, enabling communication between GH EMS and utility DR systems.
IEEE 1451 [151], [152]	Sensor Networks	Smart transducer interface with TEDS for plug and play sensors/actuators.	Ensures seamless integration of multi vendor greenhouse sensors and actuators.
IEC 61850 [153]	Substation and DER Communication	Data models and communication protocols for utility/DER automation.	Models and controls greenhouse DERs e.g. PV, storage, controllable loads
ISO 19156 [151]	Geographic Information Standards	Schema for observations and measurements.	Standardizes microclimate data exchange.
ISO 11783 (ISOBUS) [154]	Agricultural Machinery Communication	CAN based communication for agricultural machinery.	Enables interoperable automation equipment in greenhousefarm operations.
OpenADR 2.0 [155], [156]	Demand Response Automation	Protocol for automated DR with VEN/VTN roles.	Automates DR event handling for GH load shifting and energy market integration.

In another study, supplemental GH lighting was dynamically modulated in response to electricity price signals, demonstrating that energy intensive horticultural loads can be shifted or curtailed without negatively impacting crop yield or quality. Similarly, pilot projects adopting OpenADR 2.0 have shown that GHs can participate as flexible resources at the distribution grid level, providing load shifting and curtailment services with maintaining microclimate stability [156]. These examples highlight how adherence to open standards not only improves technical interoperability but also supports practical deployment of DSM strategies with measurable economic and operational benefits.

IX. GAPS, CHALLENGES, AND FUTURE RESEARCH DIRECTIONS

AI driven forecasting, DSM, and control frameworks have advanced greenhouse energy management, their large scale deployment remains limited by challenges that are unique to controlled environment agriculture. The residential or industrial energy systems, greenhouse DSM must simultaneously manage crop physiological safety, microclimate stability, and renewable driven energy variability, making many generic AI solutions insufficient. This section identifies the most critical and greenhouse specific research gaps, distilled from the reviewed literature, and highlights priority directions where future research can deliver the highest impact.

A. CROP MICROCLIMATE ENERGY OPTIMIZATION

Existing DSM approaches optimize energy consumption or cost with maintaining temperature, humidity, and CO₂ levels within fixed comfort ranges. However, crop responses to microclimate conditions are not static, they vary with growth stage, duration of exposure, and interaction among environmental variables. As a result, treating crop comfort as fixed constraints oversimplifies the biological processes involved and may lead to suboptimal or even harmful control decisions. The absence of coupled optimization frameworks that link energy management with crop growth sensitivity remains a key limitation. Integrating growth stage awareness, stress accumulation, and yield related indicators into DSM

objectives is essential for achieving biologically consistent and reliable greenhouse energy management.

B. LIMITED GENERALIZATION

AI models used for greenhouse energy forecasting and control are commonly trained using short term, site specific datasets. These models often perform well under the conditions in which they are trained, but they frequently fail when applied across different seasons, crop types, or greenhouse structures. Seasonal changes in solar radiation, ventilation behavior, and canopy development significantly alter greenhouse energy dynamics, limiting the robustness of purely data driven models. Addressing this issue requires generalization focused modeling strategies that account for seasonal transitions and crop development stages. Hybrid approaches combining physical greenhouse models with data driven learning may improve robustness, but their integration into DSM pipelines remains limited. Establishing standardized evaluation practices across seasons and crops is also necessary to assess model reliability under realistic operating conditions.

C. SELF ADAPTIVE CONTROL SYSTEMS

Greenhouse DSM systems should respond to rapid variations in weather conditions and renewable energy generation at the same time maintaining stable microclimate conditions. Although adaptive control and reinforcement learning methods have been proposed to handle such variability, their direct application is constrained by the risk of unsafe actions. In greenhouse environments, control errors can lead to irreversible crop stress or yield loss. Therefore, further emphasize risk sensitive control strategies that prioritize crop safety while adapting to uncertainty. This includes embedding operational and agronomic constraints directly into control algorithms and validating adaptive strategies using safe simulation environments. Developing control frameworks that balance adaptability with reliability remains a major challenge in intelligent greenhouse DSM.

D. INTEROPERABILITY AND SCALABILITY

Studies focus on single greenhouse systems, limiting their relevance for large scale or commercial deployments. In practice, greenhouses often operate in clusters and interact with local distribution networks, renewable generation systems, and dynamic electricity tariffs. Uncoordinated control across multiple units can reduce system efficiency and increase grid stress. The lack of scalable and interoperable DSM architectures capable of coordinated multi greenhouse operation remains a key limitation. In addition, greater attention is needed on greenhouse participation in grid level services, such as demand response and tariff based scheduling. Integrating greenhouse DSM with broader smart grid frameworks represents an important direction for improving both economic and energy system performance.

X. Conclusion

This review has critically synthesized the progression of energy focused management in smart greenhouses from early rule based scheduling and isolated control heuristics to advanced AI enabled, uncertainty aware optimization pipelines. Classical energy management strategies such as priority rules, fixed time slot planning, and deterministic actuation provide operational simplicity, but they struggle under volatile electricity tariffs, intermittent renewable generation, and constrained microclimates. Forecasting inaccuracies resulted into infeasible schedules, and these conventional methods lack mechanisms to generalize across climatic regions, seasonal variations, or crop specific environmental requirements. Recent developments demonstrate that machine learning and deep learning forecasting models particularly LSTM, GRU, CNN-LSTM, and hybrid XGBoost-LSTM-substantially improve multi horizon predictions, supporting more adaptive and cost efficient scheduling. Likewise, optimization driven DSM using MILP, MINLP, and metaheuristics enables explicit handling of operational and biological constraints, while Model Predictive Control introduces receding horizon robustness under variable conditions. Reinforcement learning further contributes autonomous policy formation, though its deployment remains limited by biological risk considerations and a lack of field scale validation. Across techno economic evaluations, this enabled PV deployment, dynamic pricing, and automation have demonstrated 17-27% reductions in operational energy cost, while improving renewable self consumption and reducing financial recovery cycles from 8-10 years to approximately 3-7 years, underscoring the commercial readiness of integrated DSM adoption. Despite these advances, key structural limitations persist, limited transferability across climate zones, uneven reporting of techno economic metrics, and scarce demonstrations of full edge computing deployment. A noticeable gap remains in quantifying renewable self consumption, payback periods,

and operational risk factors required for commercial adoption.

Future DSM research should prioritize uncertainty quantification across forecasting and control, develop lightweight models suited for embedded greenhouse hardware, incorporate domain adaptation to support geographic portability, and establish standardized economic and environmental evaluation benchmarks. Addressing these challenges will accelerate the transition from algorithm centric experimentation toward scalable, grid interactive, and commercially viable DSM frameworks capable of supporting next generation smart greenhouse energy management.

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Soumya Ranjan Biswal holds a bachelor's degree in electrical and electronic engineering from GIET, Bhubaneswar, and a master's degree in power electronics and drives from IGIT, Sarang, Odisha. He is currently a Ph.D. Research Scholar at KIIT Deemed to be University with research domains in Demand Side Management for Smart Grids. He is proficient in working with embedded devices such as Raspberry Pi and Arduino. His research

domains include smart meters, energy management, automation, IoT, and their applications.



Tanmoy Roy Choudhury (M'17, SM'20) was born in Udaipur, Tripura, India, in 1987. He received the B.E. (EEE) and M.Tech. (EE) from NIT Agartala, Tripura, in 2009 and 2012, respectively and Ph. D. from KIIT DU – Bhubaneswar in 2018. He was with JK Tyre & Industries, Ltd., during 2009–2010 and served as Assistant Professor in electrical engineering at KIIT during 2012 - 2023. He is presently

working as an Assistant Professor at NIT Rourkela. He has been selected for TARE Fellowship by SERB (Govt. of India) in 2022. His research interests include dc–dc converters, switched power supplies, renewable

energy conversion, optimal power utilization, demand side management, etc. Dr. Tanmoy is associated with different IEEE societies such as IES, IAS, PES, etc.



Babita Panda is an expert in renewable energy and photovoltaic systems. She received her B.E. from BIET, Bhadrak, and Ph.D. from KIIT DU, Bhubaneswar. She is presently working as an Associate Professor at KIIT DU. Her outstanding work has been recognized with prestigious awards, including the Vice Chancellor's Silver Medal (2009) and the Women Researcher Award (2021). Her research interests include power electronics and drives,

distributed generation, renewable energy, and Standalone and grid-connected Photovoltaic systems.



Subhrajyoti Mishra obtained her bachelor's and master's degrees from OUAT, Odisha. She obtained a Ph.D. from JAU, Junagadh, Gujarat. She was a gold medalist for her M.Sc. and Ph.D. research. Dr. Mishra is currently working in the Department of Fruit Science, IAS, SOADU, Odisha. Her research work focuses on crop phenology, seasonal influence on crop behavior, environmental impact on crop growth, standardization of vertical production systems,

precision farming and Hi-Tech farming implementing AI